

An Exact Error Theory for Mixed-Invariant Preintegration on $SE_2(3)$: Proved Truncation Residuals, a Sufficiency Bound for Deployed Coning Algorithms, and a Delayed-Fusion Error-State Filter

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[REVIEW] Author list, order, and affiliations are now verified against the published paper [1] (Li-Yu Lin, Kartik A. Pant, Benjamin Perseghetti, James Goppert; Purdue for all but Perseghetti, Rudis Laboratories). Remaining: confirm e-mail addresses, who is corresponding author, and whether authorship of this extension differs from that list. Format note: this build uses conference-mode IEEEtran with journal-style thanks blocks; decide the target template and align the author block accordingly.

Abstract—Production autopilots already accumulate inertial measurements with a first-order hold and a two-sample coning correction; PX4 and ArduPilot have shipped both for years. What the strapdown literature does not supply is an *exact* error theory for that practice. This paper provides one for mixed-invariant preintegration on the time-extended $SE_2(3)$, which is the Galilean group. For a linear generator we prove that the Magnus exponent’s T^4 grade vanishes identically and exhibit the exact T^5 residual with proved rational coefficients $-1/240$ and $-1/720$, derived from the free Lie algebra with *no* assumption on the input spectrum, where classical multi-sample coning coefficients are obtained by fitting an assumed sinusoidal input. The exact residual yields the headline result: the two-sample coning algorithm deployed in PX4 and ArduPilot is *provably adequate at flight rates*, with a computable margin of four to five orders and explicit crossovers (below 67 to 130 Hz sampling, or above 1252 rad/s body rates) beyond which it would not be. It also reframes a negative measurement, that first-order hold is marginal at the deployed 1600 Hz rate, as the predicted consequence of the same bound rather than an anomaly. The corrections enter the exponent of an exact closed form, in which gravity is a left factor, the body input a right factor, and the velocity-to-position coupling exact through a nilpotent time block, so a corrected step is still one exact flow evaluation: measured at matched accuracy it costs 284 flops per interval against 396 for a four-evaluation Runge-Kutta step, while the deployed pattern, which bolts the same corrections in front of an approximate propagation, is two orders less accurate in velocity and position once the interval is long enough for the

coupling to matter. On that foundation we build an end-to-end bound composing interpolation, truncation, and reapplication with explicit constants, in which bi-invariance and a 2-step nilpotent ideal eliminate the usual Gronwall factor on attitude; an online a posteriori certificate computed from interior sample residuals; exact reapplication of buffered preintegrals through the state-independent factorization, together with a proof that exact bias transport is impossible in any framework, which makes the second-order remainder intrinsic; a centered quadratic hold that is architecturally novel because sensor-backend accumulation is necessarily causal while delayed fusion licenses non-causal stencils (its practical value open pending measurement); and a unification in which coning, sculling, and scrolling are components of one bracket, extended consistently to the bias Jacobians of the correction terms that production preintegration does not carry. Part II documents the 15-state delayed-fusion error-state filter this theory serves and its validation, which is simulation and standalone replay throughout, never flight hardware: a 45 s mission at 0.149 m navigation RMSE and a fairness-audited stationary-pad replay at 0.076 m against 0.081 m (ArduPilot EKF3) and 0.285 m (PX4 EKF2) on identical sensor streams.

Index Terms—Lie group, mixed-invariant, preintegration, strapdown inertial navigation, Magnus expansion, coning, sculling

I. INTRODUCTION

PREINTEGRATION of inertial measurements between estimator updates is standard practice in visual-inertial and GNSS-inertial navigation. The closed-form mixed-invariant preintegration result of [1], building on the closed-form strapdown solution of [2], integrates one interval of held gyroscope and accelerometer measurements exactly on the group $SE_2(3)$: the world-frame input (gravity) acts by left multiplication, the body-frame input (angular rate and specific force) by right multiplication, and the velocity-to-position coupling is carried by a nilpotent block so that the whole flow is a product of matrix exponentials with closed-form blocks.

That result is exact for a held input. Real integrators do better than holding: they interpolate. On the flight vehicle that motivates this work, the inertial measurement unit (IMU) streams at 1600 Hz (the deployed configuration, gyro ODR 1600 Hz with a 200 Hz silicon low-pass; a 5.76 kHz high-rate

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mode was evaluated and is retained here only as a measured reference point, not as a design case), while the estimator steps at rates from 200 Hz in the reference wiring to 1 kHz in the flight-codegen study, so an estimator interval spans one to eight IMU samples which a driver-side preintegrator is intended to compose. The question this paper answers is not how to interpolate, which production autopilots settled years ago, but how large the error of doing so provably is, and what that licenses one to say about the algorithms already flying.

One architectural fact licenses everything this paper develops, and we state it up front: the estimator fuses measurements at a delayed horizon, approximately 200 ms behind the live edge, so IMU samples already exist on *both sides* of nearly every interval being integrated for the fused state. Integration for the fused state is therefore a smoothing problem, not a causal-filtering problem: a first-order hold (FOH, the linear interpolant to the interval's *endpoint* samples), and centered higher-order holds beyond it, are admissible at zero added lag, where a purely causal filter could not justify them without paying latency (Remark 6). The strictly causal consumers of the same IMU stream, above all the body-rate feedback loop, are exactly the paths that do *not* receive these holds (Remark 7). Two questions then follow: how much does ZOH cost at these rates, and can the closed form be upgraded to FOH without giving up the closed form?

The cost is not academic. The classic driver of non-commutativity error on a multirotor is frame vibration: angular vibration of amplitude ϕ on two axes 90° out of phase at frequency f produces a true coning rate of approximately $\frac{1}{2}\phi^2(2\pi f)$. At $\phi = 0.5$ mrad and $f = 200$ Hz this is roughly $0.009^\circ/\text{s}$, the same order as gyroscope bias instability. Sub-sample composition at the deployed 1600 Hz suppresses most of it; a higher-order hold removes most of the remainder for the cost of one or three Lie brackets, and, more importantly, replaces an empirical residual with a proved bound. How much of that remainder is worth removing at the deployed rate is itself a result of the crossover analysis of Section VII, not an assumption.

Prior art, and what it does not include: The *correction* this paper analyzes is already deployed; the *propagation* it corrects is not. PX4 accumulates every gyro sample trapezoidally, $\Delta\alpha = (\omega_k + \omega_{k+1})h/2$ (src/modules/sensors/Integrator.hpp, line 135), which is the first-order hold, and applies a two-sample coning correction (line 176); ArduPilot applies the same form (libraries/AP_InertialSensor/AP_InertialSensor_Backend.cpp, lines 340 to 348, citing Tian et al. [3]); sculling has the same lineage [4]. Section XVIII proves those corrections are exactly the $\mathfrak{so}(3)$ and velocity components of the bracket derived here, with the identical $1/12$ coefficient, so no novelty is claimed for the increment.

What both stacks then do with that increment is the standard strapdown pattern: rotate the attitude quaternion by the exponential of the corrected delta angle, rotate the corrected delta velocity into the world frame and add it to velocity,

add gravity times the interval separately, and advance position by a first-order update. The corrections are *bolt-ons in front of an approximate propagation*. The mixed-invariant closed form [1], [2] is structurally different: gravity is an exact left factor, the body input an exact right factor, and the velocity-to-position coupling falls out exactly through the nilpotent time block, so the coupling those updates approximate is obtained in closed form. That closed form is also *cheaper* than a Runge-Kutta step of the same problem: [1] reports a 45% reduction in floating-point operations per integration step (677 for RK-4 against 360), counted on CasADi equation graphs, with the advantage growing as the step grows because RK-4 must subdivide to hold a tolerance while the closed form does not.

Contributions: With $N(t) = N_0 + tN_1$ the linear body generator on the time-extended algebra, in order of prominence:

- 1) *the exact truncation residual:* the T^4 Magnus grade vanishes identically (Lemma 2) and the T^5 residual is $-\frac{1}{240}[N_1, [N_0, N_1]] - \frac{1}{720}[N_0, [N_0, [N_0, N_1]]]$ with both coefficients proved by two independent routes (Lemma 3), derived from the free Lie algebra with no assumption on the input spectrum, where classical multi-sample coefficients are fitted to an assumed sinusoidal coning input;
- 2) *the hold corrections unified into an exact propagation:* the closed form and its efficiency result are the prior papers' foundation [1], [2], in which gravity is an exact left factor, the body input an exact right factor, and the velocity-to-position coupling exact through the time block. What this paper adds is that the hold corrections (coning, sculling, scrolling, and the next order) enter *that* closed form as terms of the exponent (Theorem 1, (16)), at the cost of one Lie bracket, so a corrected step remains *one exact flow evaluation*. Buying the same accuracy inside a numerical integrator requires a higher-order scheme with several flow evaluations per step: measured at matched accuracy, 284 against 396 flops per interval versus a four-evaluation RK4, while the deployed bolt-on pattern, cheaper still per interval, is two orders less accurate in velocity and position once the interval is long enough for the coupling to matter (Section XVIII, Table VIII);
- 3) *the sufficiency result* (Corollary 3): because the residual is exact, the two-sample coning algorithm deployed in PX4 and ArduPilot is provably adequate at flight rates, with a computable margin of four to five orders and explicit crossovers in h and $\|\omega\|$ beyond which it is not; the same bound predicts, rather than merely records, that first-order hold is marginal at the deployed 1600 Hz rate (Section VII);
- 4) *the end-to-end bound* (Section IV): interpolation, truncation, and reapplication composed with explicit constants, in which bi-invariance and the 2-step nilpotent ideal (Proposition 2) eliminate the Gronwall exponential on attitude, making a per-second statement possible at all;
- 5) *the online a posteriori certificate* (Proposition 5): a per-interval error bound computed from interior-sample

- residuals, shippable inside the generated flight code;
- 6) *exact reapplication* (Theorem 6): the state-independent LX_0R factorization makes a horizon correction a reapplication of precomputed factors rather than a re-integration, with a proved second-order bias-relinearization remainder (Proposition 8) that is *intrinsic*, since exact bias transport is impossible in any framework (Section VI-C);
 - 7) *the centered quadratic hold* (Proposition 6), architecturally novel because sensor-backend accumulation is necessarily causal while the delayed fusion horizon licenses non-causal stencils at zero added lag (Remark 6); its practical value at the deployed rate is measured against the harness and remains open pending a flight-estimator A/B;
 - 8) *unification*: coning, sculling, and scrolling are the three components of one bracket (Proposition 2, (15)), extended consistently to the bias Jacobians of the correction terms (Proposition 4), which production preintegration does not carry; with the mixed splitting exact at every order (Lemma 4) and the sign of the bracket pinned to the composition convention (Lemma 1).

All claims carry an explicit status: *proved*, *verified numerically*, or *open*; Part II extends the vocabulary with *audited* (read directly from the named implementation source, which is Part II's ground truth), *measured* (an instrument produced the number, command trail cited), and *reported* (cited external literature or vendor source). Section VII reports the falsification experiments; Section VIII gives the flight-rate implementation and error budget.

Document role. This document is a single comprehensive development reference in two parts: Part I develops and bounds the propagation engine; Part II (Sections IX through XX) documents the delayed-fusion error-state filter that consumes it, from the implementation as ground truth. It is written to be split later along recorded lines (a journal paper, an internal architecture document, and an optional tools paper; see the comment block at the top of the source), but is maintained as one paper now. Validation throughout is simulation and standalone replay; no flight-hardware data appears anywhere in this document.

The delayed-fusion admissibility stated above is developed in Section V; its computational complement runs through the second half of the paper: the mixed-invariant factorization makes the delayed architecture viable, because when the filter corrects the state at the horizon, the buffered preintegrals are *reapplied* on top of the corrected state exactly, never re-integrated (Section VI), and the end-to-end bound of Section IV covers the full loop, interpolation plus truncation plus reapplication remainder.

Related work: The expansion is due to Magnus [5]; the linear-generator second term and the general term structure are surveyed in [6]. Bortz [7] derived the rotation-vector differential equation whose coning integral is the $\mathfrak{so}(3)$ component of our bracket. Savage [8], [4] systematized multi-sample coning and sculling algorithms; Section VIII relates

the per-step form of our correction to that lineage. Barfoot [9] discusses Magnus-based continuous-time integration on matrix Lie groups, the pointer that prompted this work. Eckenhoff, Geneva, and Huang [10] derive closed-form continuous preintegration on $SO(3) \times \mathbb{R}^n$, a comparable goal without the mixed-invariant structure or the time-block coupling that this paper exploits. The equivariant-filter line arrives at the same group from the symmetry side: Delama, Fornasier, Mahony, and Weiss [11] preintegrate on the Galilean group $\text{Gal}(3)$, which is slot for slot the time-extended $SE_2(3)$ used here, and Fornasier et al. [12] analyze bias-inclusive symmetries on its tangent group; Section VI-C makes the identification precise and shows the FOH closed form lifts to the bias-inclusive extension verbatim.

Part I: The Propagation Engine and Its Bounds

II. PRELIMINARIES

Notation follows [1], [2].

[REVIEW] Notation below is mirrored from the LaTeX source of the arXiv preprint 2310.04886 and from the reference implementation (`LieGroups.SE23` in the `modelica_models` repository). The published paper [1] has since been obtained; its results are stated for $SE_n(3)$, of which the $SE_2(3)$ used throughout here is the case $n = 2$, so this paper works in a specialization of that scope rather than a different setting. Align symbols with the published version. Note: the preprint's Theorem 1 contains apparent typographical slips relative to both the standard closed form and the reference implementation, including its C_1 and C_3 expressions, an $\Omega(C_1 I_2 + C_2 B)$ term missing the factor A that its own proof carries, an omitted P_0 in the final product, and a θ defined without the t scaling at which $\mathcal{P}$ is invoked; the forms below follow the implementation.

A. The group $SE_2(3)$ and its time-extended generators

The group of double direct isometries $SE_2(3)$ is represented as

$$X := \begin{bmatrix} R & v & p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} =: \begin{bmatrix} R & P \\ 0_{2 \times 3} & I_2 \end{bmatrix}, \quad P \equiv [v \ p] \in \mathbb{R}^{3 \times 2}, \quad (1)$$

where $R \in SO(3)$ is the attitude (body to world), $v \in \mathbb{R}^3$ the world velocity, and $p \in \mathbb{R}^3$ the world position. The algebra $\mathfrak{se}_2(3)$ consists of $[x]^\wedge = \begin{bmatrix} \Omega & U \\ 0 & 0 \end{bmatrix}$ with $\Omega = [\omega]^\wedge \in \mathfrak{so}(3)$ and $U = [u_1 \ u_2] \in \mathbb{R}^{3 \times 2}$.

The strapdown kinematics

$$\dot{R} = R[\omega]^\wedge, \quad \dot{v} = Ra + a_g, \quad \dot{p} = v, \quad (2)$$

with a the body specific force and a_g the constant gravity vector, form a *mixed-invariant* system $\dot{X} = MX + XN$ [1] with the time-extended generators

$$M := \begin{bmatrix} 0_{3 \times 3} & A_M \\ 0_{2 \times 3} & -B \end{bmatrix}, \quad N := \begin{bmatrix} \Omega & A_N \\ 0_{2 \times 3} & B \end{bmatrix}, \quad (3)$$

$$A_M := [a_g \ 0], \quad A_N := [a \ 0], \quad B := \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B^2 = 0.$$

M and N are not elements of $\mathfrak{se}_2(3)$; they live in the *time-extended* space

$$\hat{\mathbf{g}} := \left\{ \xi(\Omega, U, \beta) = \begin{bmatrix} \Omega & U \\ 0_{2 \times 3} & \beta B \end{bmatrix} \right\}, \quad (4)$$

with $\omega \in \mathbb{R}^3$, $U \in \mathbb{R}^{3 \times 2}$, $\beta \in \mathbb{R}$, which is closed under the matrix commutator (Appendix A). The nilpotent B carries the velocity-to-position coupling: right multiplication by B pushes the first column of U into the second, $UB = [0 \ u_1]$, which is exactly $\dot{p} = v$; the split of B with opposite signs between M and N keeps the bottom block of the state at I_2 since $(I - Bt)(I + Bt) = I$.

The time-extended group these generators exponentiate into (attitude, velocity, position, and a time slot with the nilpotent corner) is the Galilean group $\text{Gal}(3)$, the same group on which the equivariant preintegration line operates [11]; we keep the $SE_2(3)$ -plus-time-block notation of [1] throughout and use the $\text{Gal}(3)$ name when connecting to that literature (Section VI-C).

B. The ZOH closed form (prior result)

Proposition 1 (ZOH preintegration [1], [2]). *For constant ω , a , a_g on $[0, T]$, the solution of (2) is $X(T) = e^{MT} X(0) e^{NT}$ with*

$$e^{NT} = \begin{bmatrix} e^{\Omega T} & \mathcal{P}(\Omega T, A_N T, B T) \\ 0 & I + B T \end{bmatrix}, \quad (5)$$

$$e^{MT} = \begin{bmatrix} I_3 & \mathcal{P}(0, A_M T, -B T) \\ 0 & I - B T \end{bmatrix}, \quad (6)$$

where, with $\theta = \|\omega\| T$,

$$\begin{aligned} \mathcal{P}(\Omega, A, B) &= A + \frac{1}{2} AB + \Omega A (C_1 I_2 + C_2 B) \\ &\quad + \Omega^2 A (C_2 I_2 + C_3 B), \\ C_1 &= \frac{1 - \cos \theta}{\theta^2}, \quad C_2 = \frac{\theta - \sin \theta}{\theta^3}, \quad C_3 = \frac{\theta^2/2 + \cos \theta - 1}{\theta^4}. \end{aligned} \quad (7)$$

The body factor multiplies the state on the *right*; this fixes every sign below. Proposition 1 is implemented verbatim as the mixed exponential `exp_mixed`, the routine whose reading fixed every convention in this paper; Listing 1 shows the load-bearing lines with the paper's objects annotated, and Appendix C reproduces the full function.

C. First-order-hold input model

On one inter-sample interval $[0, T]$, model the body input as linear:

$$\begin{aligned} N(t) &= N_0 + t N_1, \\ N_0 &= \begin{bmatrix} \Omega_k & A_k \\ 0 & B \end{bmatrix}, \quad N_1 = \frac{1}{T} \begin{bmatrix} \Delta \Omega & \Delta A \\ 0 & 0 \end{bmatrix}, \end{aligned} \quad (8)$$

where $\Omega_k = [\omega_k]^\wedge$, $A_k = [a_k \ 0]$ are the sample- k measurements, $\Delta \omega = \omega_{k+1} - \omega_k$, $\Delta a = a_{k+1} - a_k$, $\Delta A = [\Delta a \ 0]$. Time flows at unit rate, so the entire B block sits in N_0 and the slope N_1 has *no* B component; this structural fact carries much of Section III. Gravity remains genuinely constant.

Rate convention. Throughout, the deployed IMU sample rate is $f_s = 1600$ Hz, $h = 1/1600$ s = 625 μ s, with the silicon gyro

```
function exp_mixed "Mixed exponential for SE_2(3) INS
propagation"
input Real X0[10] "Initial state {p0, v0, q0}";
input Real l[9] "Body tangent {u2, u1, omega}: right
factor";
input Real r[9] "World tangent: left factor";
input Real B[2,2] "Nilpotent time block {{0,dt},{0,0}}";
output Real X1[10];
algorithm
// [elided: unpack omega_l, omega_r; series branch below
0.1 rad]
// C1..C3 of Prop. 1, eq. (7):
C1 := (1.0 - cos(theta)) / theta_sq;
C2 := (theta - sin(theta)) / (theta_sq * theta);
C3 := (theta_sq/2.0 + cos(theta) - 1.0) / (theta_sq *
theta_sq);
// N_l = P(Omega, A, B): the closed form, body (right)
factor
Nl := Al + 0.5 * Al * B
+ Om * Al * (C1 * I2 + C2 * B)
+ Om2 * Al * (C2 * I2 + C3 * B);
// world (left) factor uses -B: the conjugated gravity
flow
Nr := Ar - 0.5 * Ar * B
+ Om * Ar * (C1 * I2 - C2 * B)
+ Om2 * Ar * (C2 * I2 - C3 * B);
// Rl = R_r R0 R_l: body factor composes on the RIGHT
qr0 := LieGroups.SO3.Quat.product(q_r, X0[7:10]);
ql := LieGroups.SO3.Quat.product(q_r0, q_l);
// P1 = R_r0 Nl + (R_r P0 + Nr) (I + B): the time-block
gluing
P1 := R_r0 * Nl + (R_r * P0 + Nr) * IpB;
// [elided: pack X1 from P1 columns and ql]
end exp_mixed;
```

Listing 1. The mixed exponential `exp_mixed` (abridged; elisions marked, full function in Appendix C). Annotations tie the code to Proposition 1. Source: `LieGroups/SE23/Quat/exp_mixed.mo`, `modelica-models` branch `foh-preintegration`.

low-pass at 200 Hz (review box, p. 9); this is the design point. A 5.76 kHz high-rate mode ($h \approx 174 \mu$ s) appears throughout as a measured *reference* point: it is the far end of nearly a decade in h over which the scaling laws of Section III are exercised (case 7 of the implementation study and Experiment 11 of Section VII are measured there), and it is no longer a design case. The structured case F1 of Section VII uses $T = 178 \mu$ s (5.62 kHz equivalent), the value quoted in the original design analysis, and is labeled as such.

III. MAIN RESULTS

A. Magnus expansion of the right factor

Lemma 1 (linear-generator Magnus form and its sign). *Let $Y(t)$ solve the right-composed flow $\dot{Y} = Y N(t)$, $Y(0) = I$, with $N(t) = N_0 + t N_1$. Then $Y(T) = e^{\Xi(T)}$ with*

$$\Xi(T) = T N_0 + \frac{T^2}{2} N_1 + \frac{T^3}{12} [N_0, N_1] + \Xi_{\geq 4}(T). \quad (9)$$

For the left-composed flow $\dot{Z} = N(t) Z$ the bracket coefficient is $-\frac{T^3}{12}$.

Proof. Strategy: evaluate the second Magnus term for the left convention, then map to the right convention by flow inversion. For $\dot{Z} = N(t) Z$ the Magnus series [5], [6] gives $\Xi_2 = \frac{1}{2} \int_0^T \int_0^{t_1} [N(t_1), N(t_2)] dt_2 dt_1$. For the linear generator $[N(t_1), N(t_2)] = (t_2 - t_1)[N_0, N_1]$, and $\int_0^T \int_0^{t_1} (t_2 - t_1) dt_2 dt_1 = -T^3/6$, so $\Xi_2^L = -\frac{T^3}{12} [N_0, N_1]$. If $\dot{Y} = Y N$ then $\frac{d}{dt} Y^{-1} = -N Y^{-1}$, whence $\Xi^R(N_0, N_1) =$

$-\Xi^L(-N_0, -N_1)$: Magnus terms with an even number of generator letters change sign between the conventions, odd-letter terms do not. Hence $\Xi_2^R = +\frac{T^3}{12}[N_0, N_1]$. ■

Remark 1. The reference implementation composes the body factor on the right of the state (Proposition 1), so $+\frac{T^3}{12}$ is the coefficient that applies. Numerically the opposite sign exactly doubles the trapezoid-only error (Section VII), which makes any A/B test a sharp sign check.

Lemma 2 (the T^4 grade vanishes). For the linear generator, the T^4 grade of Ξ in either convention is identically zero.

Proof. Strategy: show the only candidate bracket cancels on the simplex. Grade T^4 terms carry generator words of total weight 4 with N_0 of weight 1 and N_1 of weight 2: pure- N_0 words (all brackets vanish), $[N_1, N_1]$ -type words of multidegree $(0, 2)$ (trivially zero), and words with one N_1 and two N_0 , which arise only in the third Magnus term

$$\Xi_3 = \frac{1}{6} \iiint_{t_1 > t_2 > t_3} ([N_{t_1}, [N_{t_2}, N_{t_3}]] + [N_{t_3}, [N_{t_2}, N_{t_1}]]) dt_3 dt_2 dt_1.$$

Substituting $N_{t_i} = N_0 + t_i N_1$ and using the first simplex moments of Appendix B, the coefficient of $[N_0, [N_0, N_1]]$ is $\frac{1}{6}(\int(t_3 - t_2) + \int(t_1 - t_2)) = \frac{1}{6}(-\frac{T^4}{24} + \frac{T^4}{24}) = 0$. ■

Lemma 3 (exact T^5 residual). For the right-composed flow with linear generator,

$$\Xi(T) - \left(TN_0 + \frac{T^2}{2} N_1 + \frac{T^3}{12} [N_0, N_1] \right) = \Xi_5 T^5 + O(T^6), \quad (10)$$

$$\Xi_5 = -\frac{1}{240} [N_1, [N_0, N_1]] - \frac{1}{720} [N_0, [N_0, [N_0, N_1]]];$$

in the left convention the second coefficient is $+\frac{1}{720}$.

Proof. Strategy: derive the sign-correct Magnus differential equation for the right-composed stem, show by weight grading that it closes exactly through T^5 , and match coefficients against the complete polynomial ansatz.

(i) *The right-stem Magnus ODE.* Write $Y(t) = e^{\Xi(t)}$. The left-trivialized differential identity of the exponential map is $e^{-\Xi} \dot{Y} = \text{dexp}_{-\Xi}(\dot{\Xi})$ with $\text{dexp}_X = \sum_{k \geq 0} \text{ad}_X^k / (k+1)!$ [6]; equating to $\dot{Y} = Y N(t)$ gives $\text{dexp}_{-\Xi}(\dot{\Xi}) = N(t)$, hence

$$\begin{aligned} \dot{\Xi} &= \text{dexp}_{-\Xi}^{-1}(N(t)) = \sum_{k \geq 0} \frac{(-1)^k B_k}{k!} \text{ad}_{\Xi}^k N(t) \\ &= N + \frac{1}{2} [\Xi, N] + \frac{1}{12} [\Xi, [\Xi, N]] + 0 \cdot \text{ad}_{\Xi}^3 N \\ &\quad - \frac{1}{720} \text{ad}_{\Xi}^4 N + \dots \end{aligned} \quad (11)$$

with the Bernoulli numbers $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_3 = 0$, $B_4 = -\frac{1}{30}$. (For the left stem $\dot{Z} = N(t)Z$ one has $\dot{\Xi} = \text{dexp}_{\Xi}^{-1}(N)$, flipping the odd- k terms, consistent with Lemma 1.)

(ii) *Completeness of the ansatz.* Assign N_0 weight 1 and N_1 weight 2 (it enters only as tN_1): a Lie word of multidegree (p, q) in (N_0, N_1) contributes t^{p+2q} to Ξ . By the Witt (necklace) formula, the multidegree components of the free Lie algebra on two letters with weight at most 5 have dimensions: $(p, 0)$ and $(0, q)$ vanish for $p, q \geq 2$; $(1, 1)$, $(2, 1)$, $(1, 2)$, $(3, 1)$ are each one-dimensional, spanned by

$[N_0, N_1]$, $[N_0, [N_0, N_1]]$, $[N_1, [N_0, N_1]]$, $[N_0, [N_0, [N_0, N_1]]]$. Since dexp^{-1} is a Lie series, $\Xi(t)$ is a Lie series in N_0, N_1 , so through weight 5

$$\begin{aligned} \Xi(t) &= tN_0 + \frac{t^2}{2} N_1 + a_3 t^3 [N_0, N_1] \\ &\quad + a_4 t^4 [N_0, [N_0, N_1]] + b_5 t^5 [N_1, [N_0, N_1]] \\ &\quad + c_5 t^5 [N_0, [N_0, [N_0, N_1]]] + O(t^6) \end{aligned}$$

with universal rational coefficients: the ansatz is complete, not merely plausible (t and t^2 coefficients forced by the $k = 0$ term).

(iii) *Closure through t^5 .* $\text{ad}_{\Xi}^3 N$ has the weight-5 component $\frac{t^4}{2} [N_0, [N_0, [N_0, N_1]]]$, but its Bernoulli coefficient is $B_3 = 0$; for $k \geq 4$ the pure- N_0 brackets vanish, so $\text{ad}_{\Xi}^k N = O(t^{k+1})$, contributing only $t^{\geq 6}$ to Ξ . Hence (11) closes at the displayed terms for everything through t^5 .

(iv) *Coefficient matching.* Substituting the ansatz and reducing by antisymmetry ($[[N_0, N_1], N_1] = -[N_1, [N_0, N_1]]$, $[[N_0, N_1], [N_0, N_1]] = 0$), through t^4

$$\begin{aligned} [\Xi, N] &= \frac{t^2}{2} [N_0, N_1] - a_3 t^3 [N_0, [N_0, N_1]] \\ &\quad - t^4 a_3 [N_1, [N_0, N_1]] \\ &\quad - t^4 a_4 [N_0, [N_0, [N_0, N_1]]], \\ [\Xi, [\Xi, N]] &= \frac{t^3}{2} [N_0, [N_0, N_1]] + \frac{t^4}{4} [N_1, [N_0, N_1]] \\ &\quad - t^4 a_3 [N_0, [N_0, [N_0, N_1]]]. \end{aligned}$$

Matching (11) order by order as exact rationals:

$$\begin{aligned} t^2: \quad 3a_3 &= \frac{1}{2} \cdot \frac{1}{2} &\Rightarrow a_3 &= \frac{1}{12}, \\ t^3: \quad 4a_4 &= -\frac{a_3}{2} + \frac{1}{12} \cdot \frac{1}{2} = 0 &\Rightarrow a_4 &= 0, \\ t^4: \quad 5b_5 &= -\frac{a_3}{2} + \frac{1}{12} \cdot \frac{1}{4} = -\frac{1}{48} &\Rightarrow b_5 &= -\frac{1}{240}, \\ t^4: \quad 5c_5 &= -\frac{a_4}{2} - \frac{a_3}{12} = -\frac{1}{144} &\Rightarrow c_5 &= -\frac{1}{720}. \end{aligned}$$

The parity map $\Xi^R(N_0, N_1) = -\Xi^L(-N_0, -N_1)$ of Lemma 1 flips even-letter terms only, so the left stem carries $+\frac{1}{720}$ on the four-letter bracket and the same $-\frac{1}{240}$ on the three-letter one. ■

Remark 2 (independent confirmations). The matched values $a_3 = \frac{1}{12}$ and $a_4 = 0$ reproduce Lemma 1 and Lemma 2, whose proofs evaluate the Magnus integrals directly, and the simplex second moments of Appendix B independently yield $b_5 = -\frac{1}{240}$: two derivations with no shared steps. The graded-logarithm identification of Section VII confirms all four coefficients to fourteen digits on two seeds and both stems.

Corollary 1 (residual bound). In any submultiplicative norm,

$$\|\Xi - \Xi_{\leq 3}\| \leq \frac{T^5}{60} \|N_0\| \|N_1\|^2 + \frac{T^5}{90} \|N_0\|^3 \|N_1\| + O(T^6), \quad (12)$$

with $\Xi_{\leq 3}$ the right side of (9) without $\Xi_{\geq 4}$.

B. Exactness of the mixed splitting

Lemma 4 (the splitting commutes with any hold model). Let $\dot{X} = MX + XN(t)$ with M constant and $N(t)$ arbitrary (measurable, bounded). Then

$$X(T) = e^{MT} X(0) e^{\Xi(T)}, \quad (13)$$

with Ξ the right-convention Magnus exponent of $N(\cdot)$ alone. In particular FOH introduces no left-right cross term through the time block at $O(T^3)$ or at any other order.

Proof. Strategy: peel the left flow off by a time-dependent change of variables. Let $Z(t) = e^{-Mt}X(t)$. Since M is constant, $\dot{Z} = -Me^{-Mt}X + e^{-Mt}(MX + XN) = ZN(t)$, so $Z(T) = X(0)e^{\Xi(T)}$ by Lemma 1 and $X(T) = e^{MT}X(0)e^{\Xi(T)}$. No property of $N(t)$ beyond integrability is used. ■

C. Structure of the extended algebra

Proposition 2 (2-step nilpotency and the bracket decomposition). Write $\xi_i = \xi(\Omega_i, U_i, \beta_i) \in \hat{\mathfrak{g}}$. Then

$$[\xi_1, \xi_2] = \xi([\Omega_1, \Omega_2], \Omega_1 U_2 - \Omega_2 U_1 + \beta_2 U_1 B - \beta_1 U_2 B, 0) : \quad (14)$$

every bracket has zero B component. The translational-plus-time ideal $\hat{\mathfrak{n}} = \{\xi(0, U, \beta)\}$ is 2-step nilpotent: $[\hat{\mathfrak{n}}, [\hat{\mathfrak{n}}, \hat{\mathfrak{n}}]] = 0$, and the bracket of two ideal elements survives only in the position column, $[\xi(0, U_1, \beta_1), \xi(0, U_2, \beta_2)] = \xi(0, (\beta_2 u_{1,1} - \beta_1 u_{2,1})$ in column 2, 0). For the FOH generators (8) ($\beta_{N_0} = 1, \beta_{N_1} = 0$), the bracket $\frac{T^3}{12}[N_0, N_1] = \xi([c_\omega]^\wedge, [c_v \ c_p], 0)$ decomposes into

$$\begin{aligned} c_\omega &= \frac{T^2}{12} \omega_k \times \Delta\omega & (\text{coning}), \\ c_v &= \frac{T^2}{12} (\omega_k \times \Delta a - \Delta\omega \times a_k) & (\text{sculling}), \\ c_p &= -\frac{T^2}{12} \Delta a & (\text{scrolling}). \end{aligned} \quad (15)$$

Proof. Strategy: block multiplication, then specialize. Equation (14) is the computation of Appendix A; the B component vanishes because $B^2 = 0$ and $[\beta_1 B, \beta_2 B] = 0$. On the ideal, (14) reduces to $(\beta_2 U_1 - \beta_1 U_2)B$, which has only a position column and zero β ; a further bracket with any ideal element applies B to a position-column-only matrix (zero) or multiplies by the inner $\beta = 0$ (zero). For (15), substitute (8) into (14): the $\mathfrak{so}(3)$ part is $[\Omega_k, \Delta\Omega]/T = [\omega_k \times \Delta\omega]^\wedge/T$; the first ideal column is $(\Omega_k \Delta A - \Delta\Omega A_k)/T$; the corner term $-\beta_{N_0}(\Delta A/T)B$ places $-\Delta a/T$ in the position column. Scaling by $T^3/12$ gives (15). The $\mathfrak{so}(3)$ component is the Bortz coning increment [7]; the two ideal components are the sculling and position (scrolling) corrections of [8], [4]. Note the minus sign on the scrolling term: it is invisible to attitude-only tests and shows only in position. ■

Corollary 2 (exact termination without rotation). If $\omega_k = \Delta\omega = 0$ the FOH problem is confined to $\hat{\mathfrak{n}}$, all Magnus terms of order three and higher vanish, and $\Xi = TN_0 + \frac{T^2}{2}N_1 + \frac{T^3}{12}[N_0, N_1]$ is exact for any T .

D. The FOH closed form

Theorem 1 (first-order-hold preintegration). Let $\dot{X} = MX + XN(t)$ with M constant and $N(t) = N_0 + tN_1$ as in (8). Define the corrected body increment

$$\begin{aligned} \bar{\omega} &= T(\omega_k + \frac{1}{2}\Delta\omega) + \frac{T^2}{12} \omega_k \times \Delta\omega, \\ \bar{a} &= T(a_k + \frac{1}{2}\Delta a) + \frac{T^2}{12} (\omega_k \times \Delta a - \Delta\omega \times a_k), \\ \bar{u}_2 &= -\frac{T^2}{12} \Delta a, \end{aligned} \quad (16)$$

and $\Xi_{\leq 3} = \begin{bmatrix} [\bar{\omega}]^\wedge & [\bar{a} \ \bar{u}_2] \\ 0 & BT \end{bmatrix}$. Then

$$X(T) = e^{MT} X(0) e^{\Xi_{\leq 3}} + E(T), \quad (17)$$

where e^{MT} and $e^{\Xi_{\leq 3}}$ are given in closed form by Proposition 1 (with $\bar{u}_2$ entering the previously unused second column of the input to $\mathcal{P}$), and the error obeys $\|E(T)\| \leq \|X(0)\| e^{\|MT\|} e^{\max(\|\Xi\|, \|\Xi_{\leq 3}\|)} \|\Xi - \Xi_{\leq 3}\|$ with the exponential error bounded by (12); in particular $E(T) = O(T^5)$ and there is no T^4 error term.

Proof. Strategy: split exactly, truncate the Magnus exponent, and observe that the truncation stays inside the closed-form family. By Lemma 4, $X(T) = e^{MT}X(0)e^{\Xi(T)}$ exactly. By Lemma 1 and Proposition 2, $\Xi_{\leq 3} = TN_0 + \frac{T^2}{2}N_1 + \frac{T^3}{12}[N_0, N_1]$ evaluates to the stated $\bar{\omega}, \bar{a}, \bar{u}_2$, and, because the bracket has zero B component by (14), $\Xi_{\leq 3}$ has exactly the block shape $\begin{bmatrix} \Omega & A \\ 0 & BT \end{bmatrix}$ that Proposition 1 exponentiates in closed form. The error bound follows from $\|e^P - e^Q\| \leq \|P - Q\| e^{\max(\|P\|, \|Q\|)}$ applied to the exponent difference, Lemma 2 (no T^4 term), and Lemma 3 with Corollary 1. ■

Remark 3. Equation (16) is, in group form, the increment that production autopilots already compute: the trapezoid means are PX4's and ArduPilot's accumulated deltas and the rotational cross term is their two-sample coning correction, coefficient for coefficient (Section XVIII). No novelty is claimed for it here. What the group derivation adds is that the same single bracket also supplies the sculling and scrolling components, that its sign is pinned by the composition convention, and, in Lemma 3, the exact residual left behind, which is what Corollary 3 then turns into a statement about the deployed algorithms. The scrolling term $\bar{u}_2$ feeds the second input column, which is identically zero under ZOH.

E. Corrected relative-residual scaling

Proposition 3 (relative residual). Let $C = [N_0, N_1]$ and let the correction be $\frac{T^3}{12}C$. Then

$$\frac{\|\Xi - \Xi_{\leq 3}\|}{\left\|\frac{T^3}{12}C\right\|} \leq \frac{T^2}{20} \frac{\|[N_1, C]\|}{\|C\|} + \frac{T^2}{60} \frac{\|[N_0, [N_0, C]]\|}{\|C\|} + O(T^3), \quad (18)$$

and, since brackets confined to $\hat{\mathfrak{n}}$ vanish (Proposition 2), every surviving numerator term carries at least one rotation factor, giving, for generic directions at flight scales, the leading-order estimate

$$\frac{\text{residual}}{\text{correction}} \approx \frac{1}{10} \|\Delta\omega\| T + \frac{1}{15} (\|\omega_k\| T)^2. \quad (19)$$

The slope term $\frac{1}{10} \|\Delta\omega\| T$ dominates under strong vibration; the residual is not $O((\|\omega\| T)^2)$ relative in general. Status: (18) is proved; (19) is a leading-order estimate, verified numerically.

Proof. Strategy: divide (10) by the correction; (18) is exact bookkeeping, (19) an estimate whose omitted terms are named. Equation (18) is (10) with $T^5/240 = (T^3/12)(T^2/20)$ and $T^5/720 = (T^3/12)(T^2/60)$. For (19): in $[N_1, C]$ both corner

terms vanish ($\beta_{N_1} = \beta_C = 0$), so every term carries one rotation factor, either $\Delta\Omega/T$ acting on C (the displayed slope term) or C_Ω acting on N_1 's ΔA column (same order, omitted from the display); in $[N_0, [N_0, C]]$ the purely rotational chain carries Ω_k twice (the displayed quadratic term), but the corner map $U \mapsto -UB$ ($\beta_{N_0} = 1$) also produces terms of the shape $C_\Omega A_k B$ with a *single* rotation factor inside C_Ω , so a two-factors-per-chain counting would be false. The omitted terms are subdominant for generic directions at flight magnitudes; measured: 3.3×10^{-6} (this harness) and 2.0×10^{-6} (independent review checker) against the law's 3.8×10^{-6} at the flight case. ■

At the flight case F1 ($T = 178 \mu s$, $\|\omega\| \leq 10$ rad/s, $\|\Delta\omega\| \approx 0.2$ rad/s per sample), (19) predicts 3.8×10^{-6} relative; the measured value is 3.3×10^{-6} (Section VII). The magnitude claimed in earlier informal analysis is thus confirmed even though its stated mechanism was not.

F. Bias Jacobians

Assumption 1. *Gyroscope and accelerometer biases b_g, b_a are constant across the interval, and both endpoint samples see the same effective bias.*

[REVIEW] Assumption 1 holds through any settled LTI sensor filter with unity DC gain (a constant passes unchanged and cancels in the difference $u_{k+1} - u_k$), and through ICM-class internal delta-angle accumulation over equal windows. It fails for unsettled or time-varying filters, intra-interval bias variation (random-walk bias contributes an $O(\sigma_{rw} T^{3/2})$ slope error, far below the FOH term at these rates), or unequal sample spacing. The specific IMU part's filter chain has not been audited; confirm against the datasheet before shipping.

Proposition 4 (FOH bias sensitivities). *Under Assumption 1 the slope is bias-invariant, $\partial N_1 / \partial b = 0$, so*

$$\frac{\partial}{\partial b} \left(\frac{T^3}{12} [N_0, N_1] \right) = \frac{T^3}{12} \left[\frac{\partial N_0}{\partial b}, N_1 \right], \quad (20)$$

and the increment sensitivities of (16) are

$$\partial \bar{\omega} / \partial b_g = -TI_3 + \frac{T^2}{12} [\Delta\omega]^\wedge, \quad (21)$$

$$\partial \bar{a} / \partial b_a = -TI_3 + \frac{T^2}{12} [\Delta\omega]^\wedge, \quad (22)$$

$$\partial \bar{a} / \partial b_g = +\frac{T^2}{12} [\Delta a]^\wedge, \quad \partial \bar{u}_2 / \partial b = 0. \quad (23)$$

The rotation-channel Jacobian recursion of the ZOH closed form, $J_R^{k+1} = e^{-[\bar{\omega}]^\wedge} J_R^k - J_r(\bar{\omega}) T$ (with J_r the $SO(3)$ right Jacobian), becomes

$$J_R^{k+1} = e^{-[\bar{\omega}]^\wedge} J_R^k + J_r(\bar{\omega}) \left(-TI_3 + \frac{T^2}{12} [\Delta\omega]^\wedge \right), \quad (24)$$

i.e. one appended term $\frac{T^2}{12} J_r[\Delta\omega]^\wedge$ per step; the velocity and position channels append $\frac{T^2}{12} [\Delta a]^\wedge$ into the gyro-bias column and replace the $-TI_3$ factor per (22) in the accelerometer-bias column. The cross sensitivity (23) has no ZOH counterpart at the increment level.

Proof. Strategy: bilinearity, then the chain rule through the increment. Equation (20) is bilinearity of the bracket with

$\partial N_1 / \partial b = 0$. With $\partial(\omega_k - b_g) / \partial b_g = -I$, differentiate (16): for (21), $\partial_x(x \times \Delta\omega) = -[\Delta\omega]^\wedge$ composed with $-I$ gives $+\frac{T^2}{12} [\Delta\omega]^\wedge$; (22) and (23) are identical calculations on the sculling term; $\bar{u}_2$ depends only on the slope. Equation (24) is the chain rule through $\text{Log}(\exp(\bar{\omega}(b)))$ exactly as in the ZOH recursion, with $-TI_3$ replaced by $\partial \bar{\omega} / \partial b_g$. ■

[REVIEW] Status of Proposition 4: equations (20)–(23) are proved; (21) and the two-step composition of (24) are verified by central finite differences (maximum errors 8.1×10^{-14} and 1.2×10^{-10} relative). The composed velocity-channel recursion with the appended $R_{\text{mid}} \frac{T^2}{12} [\Delta a]^\wedge$ term was independently FD-verified in the adversarial review: 1.3×10^{-4} relative over two steps (residual accounted for by the midpoint scheme), against 7.6×10^{-2} without the term. The position channels, which this paper's harness left unexercised, were subsequently finite-difference verified in the flight-estimator implementation, which checks all five FOH bias Jacobians (Section VII, implementation validation): all channels are now exercised.

IV. END-TO-END ERROR BOUNDS

Theorem 1 bounds the integrator against the FOH input model; nothing so far bounds the model against the true inter-sample signal, so no overall guarantee yet exists. This section closes that loop: an interpolation bound, a flow-sensitivity bound whose generic Gronwall factor the algebra eliminates, a composition lemma, and the resulting per-second headline bound, followed by an online-computable a posteriori certificate.

A. Interpolation error

Theorem 2 (FOH interpolation error). *Let U be the true body input on one inter-sample interval $[0, h]$ and $\hat{U}$ its linear interpolant through the endpoint samples.*

- (a) *If $\dot{U}$ is Lipschitz with constant M_2 (equivalently $\|\ddot{U}\| \leq M_2$ a.e.), then $\sup_{[0, h]} \|U - \hat{U}\| \leq M_2 h^2 / 8$.*
- (b) *If U is Lipschitz with constant M_1 , then $\sup_{[0, h]} \|U - \hat{U}\| \leq M_1 h / 2$.*

Proof. Strategy: Peano kernel for (a), a convexity split for (b). (a) The error $e = U - \hat{U}$ vanishes at 0 and h and satisfies $e(t) = -\int_0^h G(t, s) \ddot{U}(s) ds$ with the (non-negative) Green's function of the two-point problem, and $\int_0^h G(t, s) ds = t(h-t)/2 \leq h^2/8$. (b) With $\theta = t/h$, $e(t) = (1-\theta)(U(t)-U(0)) + \theta(U(t)-U(h))$, so $\|e\| \leq (1-\theta)M_1 t + \theta M_1(h-t) = 2M_1 h \theta(1-\theta) \leq M_1 h/2$. ■

Physical parametrization. For the angular-rate channel under frame vibration at frequency f with angular displacement amplitude ϕ (the coning pair of Section I), the rate amplitude is $A_\omega = 2\pi f \phi$ and, sinusoidal worst case,

$$M_1 = (2\pi f)^2 \phi, \quad M_2 = (2\pi f)^3 \phi + j_{\text{man}}, \quad (25)$$

where j_{man} bounds the maneuver angular-rate second derivative (bounded angular jerk; small against the vibration term at motor frequencies). The sinusoid model, and with it the f^3 growth of M_2 , is meaningful only for $f \ll f_s/2$ (the deployed Nyquist is 800 Hz, the reference 2880 Hz); approaching $f_s/2$

TABLE I

A PRIORI FOH INTERPOLATION BOUND $M_2 h^2/8$ ON THE RATE CHANNEL (SINUSOIDAL WORST CASE, $h = 1/5760$ S), AND THE CENTERED-QUADRATIC BOUND OF SECTION V. UNITS: M_2 IN RAD/S³, BOUNDS IN RAD/S. GYRO BIAS-INSTABILITY FLOOR CITED IN SECTION I: $\approx 1.6 \times 10^{-4}$ RAD/S.

f [Hz]	ϕ [mrad]	M_2	FOH bound	quadratic bound
100	0.5	1.24e5	4.67e-4	2.62e-5
200	0.5	9.92e5	3.74e-3	4.19e-4
400	0.5	7.94e6	2.99e-2	6.70e-3
100	2.0	4.96e5	1.87e-3	1.05e-4
200	2.0	3.97e6	1.50e-2	1.67e-3
400	2.0	3.18e7	1.20e-1	2.68e-2

the samples alias, and case (b) with M_1 (which saturates near $2A_\omega$ per interval) is the guard. Table I evaluates the bound at the 5.76 kHz reference h ; at the deployed $h = 1/1600$ s the FOH column scales by $(5760/1600)^2 = 12.96$ and the centered-quadratic column by $(5760/1600)^3 = 46.7$, the same one-order-in- $2\pi f h$ statement as Theorem 5.

B. Flow sensitivity: a sharp split

Theorem 3 (flow sensitivity). *Let X and $\hat{X}$ solve $\dot{X} = MX + XN(t)$ and $\hat{X} = M\hat{X} + \hat{X}\hat{N}(t)$ from the same initial state, where the input difference has rotational part δ_ω and specific-force part δ_a with $\sup_{[0,h]} |\delta_\omega| = \varepsilon_\omega$, $\sup_{[0,h]} |\delta_a| = \varepsilon_a$. Then, with d the bi-invariant distance on $SO(3)$ and $\bar{a} = \sup \|a\| + \varepsilon_a$,*

$$d(R(h), \hat{R}(h)) \leq h \varepsilon_\omega, \quad (26)$$

$$|v(h) - \hat{v}(h)| \leq h \varepsilon_a + \frac{h^2}{2} \varepsilon_\omega \bar{a}, \quad (27)$$

$$|p(h) - \hat{p}(h)| \leq \frac{h^2}{2} \varepsilon_a + \frac{h^3}{6} \varepsilon_\omega \bar{a}. \quad (28)$$

No exponential (Gronwall) factor appears in any of the three.

Proof. Strategy: bi-invariance kills the Gronwall factor on the compact factor; the 2-step nilpotent ideal admits only polynomial growth. For (26), let $\Phi = R\hat{R}^T$. Then $\dot{\Phi} = R(\Omega - \hat{\Omega})\hat{R}^T = \Phi \cdot \hat{R}(\Omega - \hat{\Omega})\hat{R}^T$, an angular velocity of norm exactly $|\omega - \hat{\omega}|$ (conjugation preserves the norm on $\mathfrak{so}(3)$), so the rotation angle θ of Φ obeys $|\dot{\theta}| \leq |\omega - \hat{\omega}| \leq \varepsilon_\omega$; integrate. For (27), gravity cancels in the difference and $\dot{e}_v = Ra - \hat{R}\hat{a} = R(a - \hat{a}) + (R - \hat{R})\hat{a}$ with $\|R - \hat{R}\|_2 = 2 \sin(\theta/2) \leq \theta(t) \leq t\varepsilon_\omega$; integrate once, and once more for (28). ■

Remark 4 (why the split matters). *The generic Gronwall bound for $\dot{Y} = YU$ carries $\exp(\int \|U\|)$. Per sub-interval that factor is a harmless 1.004, but composed over one second it compounds to $e^{\|U\|} \approx e^{30}$; useless. In (26)–(28) the exponential is gone entirely, by bi-invariance on the compact factor and by Proposition 2 on the ideal; this is what makes a per-second bound possible at all.*

C. Composition and the per-second bound

Lemma 5 (error composition on $SO(3)$). *In the bi-invariant metric, $d(\prod_j R_j, \prod_j \hat{R}_j) \leq \sum_j d(R_j, \hat{R}_j)$.*

Proof. $d(AB, \hat{A}\hat{B}) \leq d(AB, \hat{A}B) + d(\hat{A}B, \hat{A}\hat{B}) = d(A, \hat{A}) + d(B, \hat{B})$ by right and left invariance; induct. The attitude blocks of both the true flow and the integrator are exactly such right-composed products (gravity carries no rotation), so the lemma applies to them. ■

Theorem 4 (per-second attitude bound). *Over one second of integration at sample interval h , with $W = \sup \|\omega\|$, $S = \sup \|\dot{\omega}\|$, $M_2 = \sup \|\ddot{\omega}\|$,*

$$E_{\text{att}}(1 \text{ s}) \leq \underbrace{\frac{M_2 h^2}{8}}_{\text{interp.}} + h^4 \underbrace{\left(\frac{S^2 W}{240} + \frac{S W^3}{720} \right)}_{\text{truncation}} + O(h^5) \text{ [rad]}. \quad (29)$$

Proof. Strategy: per sub-interval, one interpolation leg and one truncation leg; compose with Lemma 5. Interpolation: by Theorem 2(a) and (26), each sub-interval contributes at most $h \cdot M_2 h^2/8$; there are $1/h$ of them per second. Truncation: the rotation block of e^Ξ is Exp of the rotational part of Ξ exactly (block triangularity), $d(e^X, e^Y) \leq \|X - Y\|$ on $SO(3)$ by the argument of Theorem 3, and the rotational parts of the T^5 residual (Lemma 3) are bounded by $|\omega_B \times (\omega_A \times \omega_B)| \leq S^2 W$ and $|\omega_A \times (\omega_A \times (\omega_A \times \omega_B))| \leq W^3 S$ with $|\omega_B| \leq S$ by the mean value theorem; per second, multiply by $1/h$. The $O(h^5)$ collects the T^6 Magnus tail. ■

Remark 5 (honesty about tightness). *The bound (29) is phase-blind: it charges the worst-case $\|\delta_\omega\|$ with aligned sign on every sub-interval. For periodic disturbances the signed first-order contribution cancels over full periods, leaving only coning-scale drift, so the measured drift runs two to four orders below the certificate (Section VII: factors 389 to 4509 on pure coning signals). The looseness lives entirely in the interpolation leg; the Gronwall leg is already exact by Theorem 3. Status: (29) is proved; its tightness characterization is verified numerically.*

D. A posteriori certificate

Proposition 5 (online error certificate). *Let $\hat{U}$ be the single-line FOH model over one estimator interval $[0, H]$ with samples at $t_j = jh$, and let $r = \max_{j \text{ interior}} \|U(t_j) - \hat{U}(t_j)\|$ (computable online from the buffered samples by a few subtractions). If $\|\ddot{U}\| \leq M_2$ then*

$$\sup_{[0, H]} \|U - \hat{U}\| \leq r + \frac{M_2 h^2}{8}, \quad (30)$$

and hence the per-interval attitude error is at most $H(r_\omega + M_2 h^2/8)$ plus the truncation term of (29) scaled to H .

Proof. $e = U - \hat{U}$ has $\|\ddot{e}\| \leq M_2$ ($\hat{U}$ is linear) and is measured at the sample points ($e = 0$ at the interval endpoints, $\|e\| \leq r$ at interior samples). On each $[t_j, t_{j+1}]$ apply Theorem 2(a) to e against its own linear interpolant, whose magnitude is at most r . ■

The certificate is a per-step, online-computable error bound; it can be emitted alongside the generated flight code as part

of the verified-embedded-artifact pipeline, so every preintegral ships with its own error bar.

[REVIEW] Both the certificate and the FOH slope construction assume uniform, jitter-free, synchronized sample timestamps at $h = 1/f_s$. Timestamp jitter τ_j adds $S\tau_j$ to the slope and residual terms; asynchronous or dropped samples invalidate the stencil coefficients. The actual driver timestamp behavior has not been audited.

V. DELAYED FUSION AND CENTERED POLYNOMIAL HOLDS

The implemented realization of this architecture is the transported-Jacobian form documented in Part II (Section XI).

Remark 6 (admissibility of centered stencils). *The estimator fuses at a delayed horizon $t - D$ with $D \approx 200$ ms against $h \approx 174 \mu\text{s}$: every sub-interval $[t_j, t_j + h]$ integrated for the fused state has roughly $D/h \approx 1150$ later samples already in the delay buffer. The FOH stencil needs one sample beyond the interval start (u_{j+1} , i.e. zero beyond its end); the centered quadratic stencil below needs u_{j-1}, u_j, u_{j+1} , likewise zero samples beyond the interval end. Causality therefore binds only at the live edge: the state propagated to time t for control keeps the causal FOH form on its newest one or two sub-intervals, and those are re-integrated with the centered stencil as they enter the fused window. The fused state pays no lag for smoothing-grade integration.*

[REVIEW] Remark 6 assumes the delay buffer holds raw samples aligned to the integration grid (uniform sampling, no dropped samples) and that the 200 ms horizon is the operative fusion architecture; the horizon value is taken from the system design context, not read from a single implementation site. In the implementation, `retrodict` performs measurement-age alignment by the exact inverse ZOH mixed flow (negated increments and $-B$ time block); the buffered-window reapplication path is Section VI.

Remark 7 (the rate-feedback signal is not smoothed). *Angular velocity is not a filter state, and the admissibility of Remark 6 applies asymmetrically: the IMU stream has two consumers, and only the integrating one is delayed. In the audited implementation, the body-rate controller (a sampled proportional law with gyroscopic feed-forward on the fast device thread, 0.625 ms task period) consumes $\hat{\omega}(t) = \omega_{\text{meas}}(t) - \hat{b}_g$: the current gyro sample corrected by the estimator's bias state, on a strictly causal path with no smoothing horizon. The estimator contributes only the slowly varying bias, so the delayed fusion horizon D delays the bias estimate, never the rate signal, and the bias contribution to the rate error is bounded by the machinery already in place: $\sigma_{rw}\sqrt{D}$ from the random-walk model of Assumption 1 plus the per-fusion correction magnitude of Section VI, both far below the gyro noise floor at $D = 200$ ms. Causal filtering of this signal for control (vibration notching, loop shaping) is outside this paper's scope.*

[REVIEW] Remark 7 describes the audited model path: the rate task reads the estimator-republished bias-corrected gyro sample (`navigationEstimate`: measured minus bias state; consumed by the proportional body-moment law of the rate allocator). The modeled path carries no explicit anti-vibration filtering. *Hardware confirmed* (deployed flight stack `cerebri_rdd2:boards/mr_vmu_tropic.overlay`, ICM-45686 node, and `drivers/nxp_flexio_dshot`): the deployed IMU runs at gyro ODR 1600 Hz with the silicon gyro low-pass configured at $\text{ODR}/8 = 200$ Hz bandwidth (accelerometer $\text{ODR}/16 = 100$ Hz); the rate loop is paced directly by the IMU data-ready interrupt at 1600 Hz (matching the modeled 0.625 ms task period) and writes `DShot600` once per tick, with `imu_to_motor_latency_us` instrumented; the overlay's own comment states the filter intent (conservative enough to cut motor vibration without extra control-path threading). Anti-alias filtering therefore exists in deployment, in silicon, before sampling: the model's unfiltered path is the idealization of an already-filtered stream. Observation, not applied: the silicon LFP attenuates the coning drive *before* sampling, so the error-budget table may credit it; the budget numbers here are not rescaled. The `DShot` driver supports the telemetry request bit but not bidirectional eRPM decode, so RPM-synced notching remains future driver work.

A. Centered quadratic hold

Proposition 6 (centered quadratic hold, closed form). *On $[0, h]$ let*

$$N(t) = A + tB + t^2C, \quad B = \frac{u_{j+1} - u_{j-1}}{2h}, \quad C = \frac{u_{j+1} - 2u_j + u_{j-1}}{2h^2}, \quad (31)$$

with $A = u_j$: the interpolating parabola through the three centered samples (B, C carry no time block; all of it sits in A). For the right-composed flow,

$$\begin{aligned} \Xi &= hA + \frac{h^2}{2}B + \frac{h^3}{3}C + \frac{h^3}{12}[A, B] \\ &\quad + \frac{h^4}{12}[A, C] + \frac{h^5}{60}[B, C] + \Xi_{\text{res}}, \\ \Xi_{\text{res}} &= -\frac{h^5}{240}[B, [A, B]] - \frac{h^5}{720}[A, [A, [A, B]]] \\ &\quad + \frac{h^5}{360}[A, [A, C]] + O(h^6). \end{aligned} \quad (32)$$

Every bracket has zero B -block, so e^Ξ retains the closed form of Proposition 1; brackets confined to the ideal vanish (Proposition 2), so Ξ_{res} passes through $\mathfrak{so}(3)$ and the series terminates at the displayed Ξ terms when $\omega \equiv 0$; and $C = 0$ recovers Theorem 1 exactly.

Proof. Strategy: the same apparatus as Lemma 1 and Lemma 3, with one more letter in the grading. The pairwise commutator is $[N(t_2), N(t_1)] = (t_1 - t_2)[A, B] + (t_1^2 - t_2^2)[A, C] + t_1 t_2(t_1 - t_2)[B, C]$, and the ordered double integrals are $h^3/6, h^4/6, h^5/30$; halving them (right-stem sign per Lemma 1) gives the three Ξ_2 terms in (32). Grading: weights $(A, B, C) = (1, 2, 3)$; through weight 5 the only additional free-Lie components involving C are $[A, C]$ (weight 4), $[B, C]$ (weight 5), both already in Ξ_2 , and $[A, [A, C]]$ (weight 5), each of Witt dimension one. The $[A, [A, C]]$ coefficient comes from the Ξ_3 simplex with second moments $\int t_1^2 = \frac{h^5}{10}, \int t_2^2 = \frac{h^5}{20}, \int t_3^2 = \frac{h^5}{60} \cdot \frac{1}{6}[(\frac{h^5}{60} - \frac{h^5}{20}) + (\frac{h^5}{10} - \frac{h^5}{20})] = \frac{h^5}{360}$, with

an odd letter count, hence stem-independent. The (A, B) -only terms are unchanged from Lemma 3. ■

Proposition 7 (bias invariance of the centered differences). *A constant bias b is annihilated by both centered differences $((+1, -1)$ and $(+1, -2, +1)$ both sum to zero), so $\partial B/\partial b = \partial C/\partial b = 0$ and, by bilinearity, the bias sensitivity of (32) acts only through A : the rotation-channel increment Jacobian becomes*

$$\frac{\partial \bar{\theta}}{\partial b_g} = -hI_3 + \frac{h^3}{12}[\omega_B]^\wedge + \frac{h^4}{12}[\omega_C]^\wedge, \quad (33)$$

the same structure as Proposition 4 with one additional cross term per bracket containing A ; the $[B, C]$ term is fully bias-invariant, and the recursion shape of (24) is unchanged. Verified by finite differences to 1.7×10^{-13} .

B. Interpolation gain, noise, and the honest trade

Theorem 5 (quadratic interpolation error). *With nodes $-h, 0, h$ and evaluation on $[0, h]$: if $\|\ddot{U}\| \leq M_3$, then*

$$\sup_{[0, h]} \|U - \hat{U}_2\| \leq \frac{\sqrt{3}}{27} M_3 h^3 \approx 0.0642 M_3 h^3. \quad (34)$$

Sinusoidal worst case $M_3 = (2\pi f)^4 \phi$ on the rate channel; the gain over FOH is

$$\frac{(\sqrt{3}/27)M_3 h^3}{M_2 h^2/8} = \frac{8\sqrt{3}}{27} (2\pi f h) \approx 0.513 (2\pi f h), \quad (35)$$

one full order in $2\pi f h$: at the deployed $h = 1/1600$ s this is 0.20 at 100 Hz and 0.40 at the 200 Hz filter corner (0.056 and 0.11 respectively at the 5.76 kHz reference; Table I, last column).

Proof. Newton remainder $e = \frac{1}{6}\ddot{U}(\xi)(t+h)t(t-h)$; $\max_{[0, h]} t(h^2 - t^2) = 2h^3/(3\sqrt{3})$ at $t = h/\sqrt{3}$. ■

Noise amplification (honesty term). With white sample noise of variance σ^2 , the difference estimators amplify: $\text{Var}(B_{\text{FOH}}) = 2\sigma^2/h^2$, $\text{Var}(B_{\text{centered}}) = \sigma^2/(2h^2)$ (four times smaller), $\text{Var}(C) = 3\sigma^2/(2h^4)$. What reaches the integral is milder: the per-interval increment $\int_0^h \hat{U} dt$ has noise variance $\frac{1}{2}h^2\sigma^2$ (FOH, trapezoid weights) versus $\frac{5}{8}h^2\sigma^2$ (centered quadratic, weights $\frac{2}{3}, \frac{5}{12}, -\frac{1}{12}$), a 12% noise standard deviation penalty. The higher order therefore stops paying when the deterministic gain is smaller than the added noise, roughly when $\sigma \gtrsim 1.5 M_2 h$; below that crossover the quadratic wins on both counts. *Live edge:* the newest sub-interval of the live propagation stays FOH; its one-interval excess $h(M_2 h^2/8 - (\sqrt{3}/27)M_3 h^3)$ enters the live state once, is not cumulative, and is removed when the interval enters the fused window. *Aliasing guard:* near $f_s/2$ the polynomial model degrades and Theorem 2(b) with M_1 remains the guard, exactly as for FOH.

Certificate update. With the interior samples now checking the quadratic fit over $[0, H]$ (residual r_2), divided-difference mean value gives $2C = \ddot{U}(\eta)$ for some η in the node hull, so $\|\ddot{e}\| \leq M_3 H$ and $\sup \|U - \hat{U}_2\| \leq r_2 + \frac{h^2}{8} M_3 H$: the same online form, one order higher.

VI. RETRODICTION: EXACT REAPPLICATION AT THE DELAYED HORIZON

When the filter corrects the state at the delayed horizon, the buffered preintegrals must be applied on top of the corrected state. The mixed-invariant structure makes that reapplication exact and free.

Theorem 6 (exact reapplication). *Under $\dot{X} = MX + XN(t)$ with M constant, the flow over $[0, T]$ factors as*

$$X(T) = L X(0) R, \quad L = e^{MT}, \quad R = e^{\Xi(T)}, \quad (36)$$

with L and R independent of $X(0)$ (Lemma 4). Hence a horizon correction $X_0 \mapsto X'_0 = X_0 e^{[\delta]^\wedge}$ (right injection, as implemented) reapplies the same precomputed factors exactly, $X'(T) = L X'_0 R$, with no re-integration; and the FOH and centered-hold corrections of Theorems 1 and Proposition 6 live entirely inside R and inherit the property. In the implementation's composition order this is precisely the preintegrated prediction

$$\begin{aligned} q^+ &= q_0 \otimes \Delta q, \\ v^+ &= v_0 + g \Delta t + R_0 \Delta v, \\ p^+ &= p_0 + v_0 \Delta t + \frac{1}{2} g \Delta t^2 + R_0 \Delta p, \end{aligned} \quad (37)$$

whose gravity terms are exactly the blocks of $L = e^{M\Delta t}$ and whose $(\Delta q, \Delta v, \Delta p)$ triple is R .

Proof. Lemma 4 with $Z = e^{-Mt} X$: $Z(T) = X(0) e^{\Xi(T)}$ holds for every initial value, so L and R are state-independent by construction. That (37) equals $L X_0 R$ is the block multiplication of Section II: the world factor with the $-B$ construction contributes $g\Delta t$ to velocity and $\frac{1}{2}g\Delta t^2$ to position after the time-block coupling $p^+ \ni v_0 \Delta t$. Verified numerically: with a right-injected pose correction on the initial state, the reapplied composition matches the re-integrated mixed flow to 1.2×10^{-14} (Section VII). ■

A. Bias relinearization remainder

The correction also updates the bias estimate; R was integrated at the old linearization and is moved to the new one at first order, $R \mapsto R e^{[J \text{ db}]^\wedge}$ (rotation) with additive $J_v \text{ db}$, $J_p \text{ db}$ (translation), using the Jacobians of Proposition 4 including the FOH terms.

Proposition 8 (second-order remainder). *Let $R(b)$ be the composed rotation preintegral over a window of length T_{tot} built from per-interval increments affine in the gyro bias (as all holds above are), and let J be its exact first derivative at b_0 . Then for $\theta_{\text{rem}} = d(R(b_0 + \text{db}), R(b_0) e^{[J \text{ db}]^\wedge})$,*

$$\begin{aligned} \theta_{\text{rem}} &\leq (T_D \|db_g\|)^2, \\ T_D &= \sum_j \|D_j\| \leq T_{\text{tot}} (1 + \frac{h}{12} \sup |\Delta\omega|), \end{aligned} \quad (38)$$

and the velocity channel obeys $|\Delta v\text{-remainder}| \leq \frac{1}{2} \sup \|a\| T_{\text{tot}}^3 \|db_g\|^2$; the accelerometer-bias channel alone is exactly affine (zero remainder), and mixed terms are of order $T_{\text{tot}}^2 \|db_g\| \|db_a\|$ times h -small factors.

Proof. Strategy: bound the curvature of the product curve by bi-invariance. For $s \mapsto R(b_0 + su)$, $\|u\| = 1$, the body velocity is $V(s) = \sum_j \text{Ad}_{Q_j(s)^{-1}}(J_r(\theta_j(s))D_j u)$ with $\|J_r\| \leq 1$ and Ad norm-preserving, so $\|V\| \leq T_D$; differentiating, the Ad terms contribute commutators against tail velocities of norm at most T_D and the J_r derivative at most $\frac{1}{2}\|D_j\|$ each, giving $\|V'\| \leq 2T_D^2(1 + O(h))$. By the bi-invariant comparison of Theorem 3, $d(R(b_0 + db), R(b_0)e^{\|db\|V(0)}) \leq \frac{1}{2}\|db\|^2 \sup \|V'\| \leq (T_D\|db\|)^2$, and $\|db\|V(0) = J db$ by definition of the derivative. The velocity bound iterates the same argument through $\Delta v = \sum_j (\text{rotations}) \cdot (\text{affine increments})$ with two rotation derivatives; affinity in b_a alone is immediate since the rotations do not involve b_a and the increments are affine in it. ■

Measured (Section VII): remainder slope 1.992 against theory 2, and constant 0.063 of the bound (38), over $\|db_g\|$ from 10^{-5} to 10^{-2} rad/s on a $T_{\text{tot}} = 0.2$ s window. The end-to-end bound of Theorem 4 therefore gains a reapplication term $N_{\text{fus}}(T_{\text{tot}}\|db_g\|)^2$ per second; at the assumed typical correction below it evaluates to 4×10^{-7} rad/s worst case (2.5×10^{-8} with the measured constant), negligible against both legs of (29).

[REVIEW] Typical per-fusion correction magnitudes are assumed here as $\|db_g\| \approx 10^{-4}$ rad/s and $\|db_a\| \approx 10^{-3}$ m/s² at a 1 kHz fusion rate; these were not grounded in the implementation's tuning files and should be replaced with values from the flight logs or the covariance tuning.

B. Error-state convention consistency

Read directly from the implementation routines: the correction-injection routine `inject`, the preintegrated prediction `predictPreintegrated`, the covariance reset `conjugateReset`, the bias recorection `correctPreintegratedImu`, and the measurement-age alignment `retrodict`. Injection is right-multiplicative ($X' = X e^{\delta}$, biases additive); the preintegral is applied on the right of the state (37); the bias correction of the stored delta is right-multiplicative on the delta ($\Delta q \otimes e^{J db}$); the error-state transition of the preintegrated prediction transports the right-tangent error by the adjoint-type blocks of the delta (the ΔR^T diagonal blocks, the $-\Delta R^T[\Delta p]^\wedge$ and $-\Delta R^T[\Delta v]^\wedge$ cross blocks, and the $\Delta R^T \Delta t$ time-block coupling); and the covariance reset conjugates by $\text{diag}(J, I)$ with J the $SE_2(3)$ right Jacobian of the applied correction. All four sit on the same (right) side as the factorization of Theorem 6: *no convention disagreement was found*. A right-injected correction δ at the horizon transports through the reapplied factor as $\text{Ad}_{R^{-1}}\delta$, which is exactly the transition's adjoint block structure, so the covariance transport is the correct conjugation for reapplication through R .

[REVIEW] The consistency statement above is read from the ESKF path; the equivalence of the transition blocks with $\text{Ad}_{R^{-1}}$ on the time-extended group was verified by block inspection, not by a separate mechanized proof. The UKF path (the counterpart routines `UKF/injectVector` and

`UKF/predictPreintegrated`) has since been audited by the implementation review: right injection $X e^{\delta}$, right-tangent error log, reapplication in the form of Theorem 6, and bias hypotheses moved through the stored first-order Jacobians; no convention disagreement was found, and the sigma-point bias coupling is linearized, consistent with the second-order remainder bound of Proposition 8. Remaining item: only the block-inspection status of the Ad-equivalence.

C. Relation to equivariant preintegration on $\text{Gal}(3)$

The equivariant preintegration of [11] operates on the Galilean group, and the identification with this paper's objects is exact, slot for slot: their (A, a, b, c) is our (R, v, p, s) , their algebra $(\omega, \nu, \rho, \alpha)$ is our $(\omega, a\text{-column}, u\text{-column}, \tau)$ with the same nilpotent corner; their preintegration recursion is our right body factor; and their gravity factor Γ_{ij} , kept outside the symmetry, is exactly our world factor e^{MT} with the $-B$ construction. Both lines discovered independently that the time slot makes gravity and the velocity-position coupling exactly integrable (status: proved by direct comparison of the matrix forms). Their bias-inclusive symmetry is the tangent group $\text{Gal}(3) \ltimes \mathfrak{gal}(3)$, whose algebra has bracket $[(x_1, y_1), (x_2, y_2)] = ([x_1, x_2], \text{ad}_{x_1} y_2 - \text{ad}_{x_2} y_1)$ with an abelian fiber.

Remark 8 (the FOH closed form lifts to the bias-inclusive tangent group). *On $\text{Gal}(3) \ltimes \mathfrak{gal}(3)$ the lifted translational-plus-time ideal (whose fiber must carry the full algebra, the gyro bias sitting in the fiber's rotation slot) is 3-step, not 2-step, nilpotent: the triple bracket $[m_3, [m_1, m_2]]$ survives in the fiber, with obstruction terms $\beta_{w,3} \times (\tau_2 a_1 - \tau_1 a_2)$ (gyro-bias fiber crossed into the corner) and $\tau_3(\beta_{w,1} \times a_2 - \beta_{w,2} \times a_1)$ pushed through E ; it vanishes exactly when the gyro-bias fiber does, depth 4 dies, and for $\omega \equiv 0$ the Magnus series terminates exactly at grade 5 with $\Xi = TA + \frac{T^2}{2}B + \frac{T^3}{12}[A, B] - \frac{T^5}{240}[B, [A, B]]$ (proved in the companion analysis; the universal coefficients of Lemmas 1–3 hold on any Lie algebra and were re-identified numerically on the 10×10 realization, grade-5 projection residual 5.9×10^{-15}). None of this costs the construction anything, because the equivariant lift's fiber is algebraically slaved to the group component: for constant bias b the lifted flow with generator $(N(t), -\text{ad}_{N(t)} b)$ satisfies $X(t) = (C(t), b - \text{Ad}_{C(t)} b)$ with $\dot{C} = CN$, for any hold model, since $\frac{d}{dt}(b - \text{Ad}_C b) = -\text{Ad}_C \text{ad}_N b$ matches the fiber equation. The FOH and centered-hold closed forms and their residuals therefore lift verbatim, with zero additional truncation (verified numerically: fiber-slaving residual 3.2×10^{-16} , the RK4 floor). Statuses: nilpotency class and slaving proved; numerics verified.*

Two consequences frame the connection from strength. First, the preintegration formulation of [11] keeps gravity outside the symmetry, so the state-independent-factors property that Theorem 6 exploits holds there verbatim; the full tangent-group filter lift of [12], by contrast, conjugates gravity by the current state and would not support the buffered-reapplication architecture. Second, exact bias transport is impossible in *any* framework: $\partial Y / \partial b$ is path dependent at fixed preintegral (two

input paths with identical $\Upsilon(b)$ exhibit derivative matrices differing by 2.3×10^{-3} relative), and the equivariant line likewise performs its bias update by first-order approximation; the $O(\|db\|^2)$ remainder of Proposition 8 is intrinsic, not an artifact of this paper’s parametrization.

[REVIEW] The tangent-group numerics quoted in this subsection (slaving residual, 10×10 coefficient re-identification, the path-dependence counterexample) come from the companion equivariance analysis harness (seed 20260825), which lives in a session scratchpad; fold its eqf binary source into foh-check before publication so the results are reproducible from the repository.

D. The covariance side, bounded

Four statements delimit what FOH changes for the filter covariance; the full predict/correct treatment is deliberately out of scope (see the review box below).

(i) *Transition.* The error-state transition over one interval is a function of the composed delta, not of the raw samples: for the right-tangent error it is the adjoint-type transport

$$\Phi_{\text{nav}} = \text{Ad}_{R^{-1}}, \quad R = e^{\Xi \leq 3}, \quad (39)$$

(the ΔR^T , $-\Delta R^T[\Delta v]^\wedge$, $-\Delta R^T[\Delta p]^\wedge$, and time-block Δt blocks of the preintegrated prediction), so the transition picks up the $\frac{T^3}{12}$ bracket corrections of (16) automatically, at the same order as the mean: the implementation constructs it from the corrected delta triple and the FOH bias Jacobians, with no separate covariance-side change to make. Status: audited (the transition-block construction) plus Theorem 6’s block identification; no new derivation.

(ii) *Trapezoid noise correlation.* White sample noise of variance σ_s^2 enters the FOH increment through the trapezoid weights $h(n_k + n_{k+1})/2$, so adjacent intervals share an endpoint sample: per-interval variance $\frac{1}{2}h^2\sigma_s^2$ and one-step cross-covariance $\frac{1}{4}h^2\sigma_s^2$, i.e. correlation coefficient exactly $\frac{1}{2}$ (ZOH increments are white at $h^2\sigma_s^2$). The telescoped sum $\sum_k h(n_k + n_{k+1})/2 = h(\frac{1}{2}n_0 + n_1 + \dots + n_{K-1} + \frac{1}{2}n_K)$ has variance $(K - \frac{1}{2})h^2\sigma_s^2$: keeping the ZOH-level white model in Q_d reproduces the *accumulated* covariance exactly (to one half-sample in K) while overstating single-interval noise by a factor of two. Dropping the off-diagonal term is therefore admissible exactly when no measurement update resolves an individual sub-interval,

$$H \geq 2h, \quad (40)$$

which holds by construction (eight samples per tick at 1600 Hz with the 200 Hz reference tick); equivalently, at sub-interval update cadence the neglected term is $\frac{1}{4}h^2\sigma_s^2$ against the tuned per-interval process noise $Q_c h$, negligible whenever the tuning margin exceeds $\frac{1}{4}\sigma_s^2 h/Q_c$. Status: proved (elementary computation).

(iii) *Consistency at second order.* A covariance-model perturbation δP perturbs the Kalman gain at first order, but the gain perturbation multiplies the zero-mean innovation, so the estimate’s *mean* error is unchanged at first order and the consistency statistics (NEES) shift only by $O(\|\delta P\|/\|P\|)$; a mean-model error, by contrast, biases the innovation directly

TABLE II
MAGNUS GRADES FOR THE LINEAR GENERATOR, AS PROVED IN LEMMAS 1–3, CONFIRMED NUMERICALLY BY THE GRADED LOGARITHM (GENERIC 6×6 , TWO SEEDS; RESIDUALS $< 5 \times 10^{-14}$).

grade	$\dot{Y} = YN$ (implemented)	$\dot{Z} = NZ$
T	N_0	N_0
T^2	$\frac{1}{2}N_1$	$\frac{1}{2}N_1$
T^3	$+\frac{1}{12}[N_0, N_1]$	$-\frac{1}{12}[N_0, N_1]$
T^4	0	0
T^5	$-\frac{1}{240}[N_1, [N_0, N_1]]$ $-\frac{1}{720}[N_0, [N_0, [N_0, N_1]]]$	$-\frac{1}{240}[N_1, [N_0, N_1]]$ $+\frac{1}{720}[N_0, [N_0, [N_0, N_1]]]$

at first order. Removing the $O(T^3)$ mean error while leaving the covariance side with at most the factor-two single-interval noise shape of (ii), which telescopes away at the tick scale, therefore cannot degrade consistency beyond that bounded, sub-tick-only perturbation. Status: derived from standard EKF perturbation arguments; stated as a bound, not separately verified numerically.

(iv) *The covariance path was already higher order.* The audited implementation integrates the driven noise with Simpson quadrature over the interval, evaluating the transition at half and full steps (ESKF/discreteProcessCovariance.mo under Estimation/StrapdownINS); the covariance path was integrated to higher order than the ZOH mean path even before this work; with FOH the mean has caught up.

[REVIEW] Deliberate omissions of this subsection: the exact colored-noise recursion for the shared-endpoint correlation (the 2×2 weight structure propagated through the Jacobians rather than bounded), and the full predict/correct covariance treatment. Both are addressed in a companion architecture document in preparation.

VII. NUMERICAL VALIDATION

Every derivation above was subjected to falsification against RK4 ground truth; no experiment contradicted a derivation. Errors are Frobenius norms on the 5×5 (structured) or 6×6 (generic) matrix flow.

Coefficient identification: The flow of $\dot{Y} = YN(t)$ has the exact Taylor recursion $nY_n = Y_{n-1}N_0 + Y_{n-2}N_1$; the graded logarithm of $\sum Y_n T^n$ isolates each Magnus grade without any integrator error. On generic random 6×6 matrices (seed 42, repeated with seed 20260824), the identified grades are shown in Table II; residuals against the stated closed forms are at machine precision, the grade-4 component vanishes to 1.4×10^{-15} , and the grade-5 projection onto the two-bracket basis leaves a relative residual of 3.8×10^{-14} , confirming the coefficients proved in Lemma 3 and the completeness of the basis.

Structured flight cases: Table XI (appendix) evaluates the structured 5×5 algebra at flight magnitudes ($\|\omega\| = 10$ rad/s, $\|a\| = 20$ m/s²; per-sample deltas as listed) against RK4 truth at 10^4 substeps. The FOH correction reduces the error by four to six orders; the wrong bracket sign exactly doubles the trapezoid error; the measured error matches the T^5 prediction

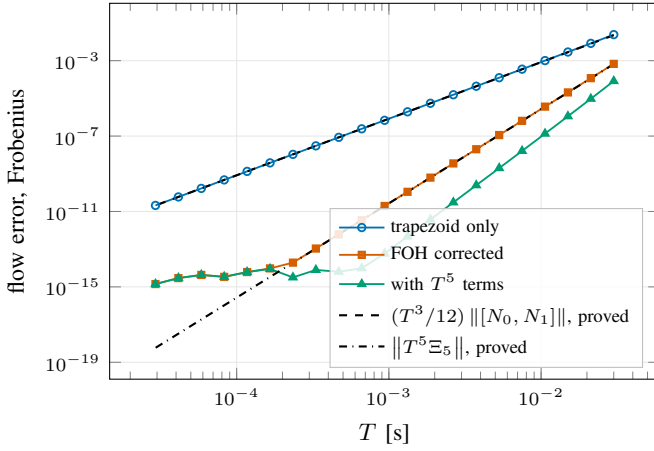


Fig. 1. Convergence sweep (Frobenius flow error vs interval length, fixed physical generators). The dashed and dash-dotted references are the proved coefficients of Lemmas 1 and 3, not fits; measured slopes are 3.006 (trapezoid only) and 5.008 (FOH corrected) against theoretical orders 3 and 5. The curve with the T^5 terms included drops to the f64/RK4 floor.

of Lemma 3 to within 0.2–3.5%; and adding the T^5 terms to the exponent drops the error by a further two orders (the T^6 tail). Precision caveat for F1: its absolute error (8.25×10^{-14}) sits within two orders of the f64/RK4 floor, so its ratio is resolved only to a few percent; an independent series-Taylor truth (adversarial review checker) puts it at 1.0008. The difference between the trapezoid-only and FOH flows matches $\left\| \frac{T^3}{12} [N_0, N_1] \right\|$ to 2.8×10^{-5} relative in case F1.

Order verification: Sweeping $T = 3 \times 10^{-2} \cdot 2^{-k/2}$, $k = 0.20$, with fixed physical N_0 ($\|\omega\| = 10$ rad/s, $\|a\| = 20$ m/s²) and N_1 (500 rad/s², 1000 m/s³), Fig. 1 overlays the measured errors on the *proved-constant* reference curves $(T^3/12) \|[N_0, N_1]\|$ and $\|T^5 \Xi_5\|$ drawn from Lemmas 1 and 3 (references, not fits). Fitted log-log slopes: 3.006 for the trapezoid-only error over all 21 points, 5.008 for the FOH error over the 11 points above 10^{-11} (kept two orders above the f64/RK4 floor of about 6×10^{-15}); the pointwise ratio of FOH error to the T^5 prediction is 1.000–1.035 on the fit set, and the curve with the T^5 terms included falls to the floor at small T .

Structural checks: Random ideal elements give exact zeros (in f64, structurally zero products) for the off-corner entries of $[\hat{n}, \hat{n}]$, for the corner formula error, and for $[\hat{n}, [\hat{n}, \hat{n}]]$; with $\omega \equiv 0$ the truncated closed form matches RK4 to 3.0×10^{-15} at $T = 3 \times 10^{-2}$ (Corollary 2). For Lemma 4, the mixed flow with gravity on matches $e^{MT} X_0 \cdot (\text{body flow})$ to 3.2×10^{-14} , and the error of the truncated split equals the body-only truncation error to four digits (1.765×10^{-10} both): the gluing contributes nothing. The ZOH closed form of Proposition 1 itself matches RK4 to 2.4×10^{-14} , and the block-series construction of $\mathcal{P}$ matches the generic matrix exponential to 8.7×10^{-19} .

End-to-end bounds and hold comparison: On seeded band-limited coning signals (coning pair of amplitude ϕ at f , $W_0 = 10$ rad/s maneuver rate, 20 m/s² specific force with 2 m/s² vibration; truth by RK4 at 24 substeps per sample

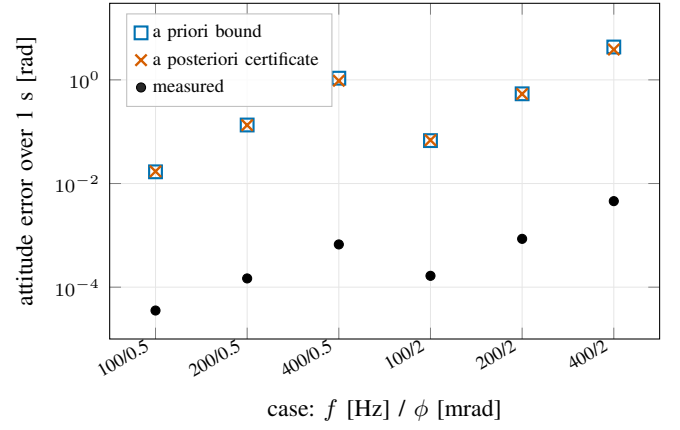


Fig. 2. Bound tightness per coning case, per-tick single-line form (the object Proposition 5 certifies; harness data, `data/tightness.dat`). The certificate always bounds the measured error; on these worst-case-achieving sinusoids it necessarily sits at the a priori level (within 11%), while the measured drift lies two to four orders below both by phase cancellation (Remark 5). On quieter-than-spec signals the certificate’s residual r shrinks with the actual deviation; the a priori bound cannot.

TABLE III
MEASURED ATTITUDE ERROR OVER 1 S, THREE HOLDS, DEPLOYED RATE AGAINST THE HIGH-RATE REFERENCE (EXPERIMENT 11).

f_s	f/ϕ	$2\pi fh$	ZOH	FOH	centered
5760	200/0.5	0.22	1.06e-4	4.04e-6	5.01e-7
5760	200/2.0	0.22	4.22e-4	2.52e-5	2.28e-7
1600	100/0.5	0.39	1.92e-4	1.27e-5	1.13e-6
1600	200/0.5	0.79	3.83e-4	5.26e-5	1.10e-5
1600	200/2.0	0.79	1.54e-3	3.20e-4	5.24e-5
1600	400/0.5	1.57	7.86e-4	2.34e-4	1.25e-4

interval over a 1 s horizon), Fig. 2 and Table XII (appendix) report the a priori bounds of Theorems 4 and 5 against measured per-second attitude error for the per-sample FOH and centered-quadratic integrators. The bounds hold in every case, with phase-blind tightness factors 389 to 4509 (Remark 5); the measured centered-over-FOH gain is $2.5\times$ to $59\times$ (phase-dependent), against bound-level gains of $4.5\times$ to $18\times$. The measured FOH error crosses the 1.6×10^{-4} rad/s bias-instability floor only at 400 Hz/2 mrad (1.69×10^{-4}), where the centered hold sits $20\times$ below it. For the single-line per-tick form, the a posteriori certificate of Proposition 5 evaluates within 11% of the a priori bound on these worst-case-achieving sinusoids (its value shows on non-worst-case signals, where r shrinks with the actual deviation) and always bounds the measured error (factors 414 to 1439); Fig. 2 shows all three quantities per case.

Hold comparison at the deployed rate: Experiment 11 integrates the same class of band-limited coning signal for 1 s at both rates and compares three holds against RK4 truth (attitude angle error, held-interval-start ZOH; seeded, deterministic phases):

Reading Table III: against this held-start baseline FOH still gains at the deployed point ($0.137\times$ and $0.208\times$ at $2\pi fh = 0.79$) but the gain has eroded by an order from the

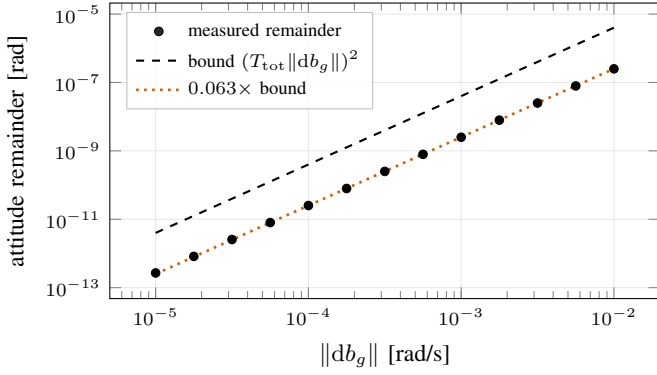


Fig. 3. Bias-relinearization remainder of the reapplied preintegral vs correction magnitude, $T_{\text{tot}} = 0.2$ s window (data/remainder.dat). Measured slope 1.992 against theory 2; the points sit at a constant 0.063 of the proved bound (38).

5.76 kHz reference ($0.038\times$, $0.060\times$), and at $2\pi fh = 1.57$ it has eroded to $0.30\times$; the centered quadratic hold is a further $0.16\times$ to $0.21\times$ below FOH at the deployed point. The erosion with $2\pi fh$ is the crossover of Fig. 5 seen from the other ZOH variant: the held-start baseline has no phase-lag term for FOH to beat, so FOH never loses outright here, but the margin collapses in exactly the regime the bound predicts. All measured.

Reapplication: On a $T_{\text{tot}} = 0.2$ s window with band-limited inputs, the reapplied preintegral with exact first-order bias Jacobians was compared against full re-integration at the corrected bias: the attitude remainder fits slope 1.992 against theory 2 over $\|db_g\| \in [10^{-5}, 10^{-2}]$ rad/s, at 0.063 of the bound (38) (Fig. 3; e.g. 2.5×10^{-11} rad at 10^{-4} rad/s); the velocity remainder is likewise quadratic and within its bound.

With a right-injected pose correction on the initial state, the reapplied factorization matches the re-integrated mixed flow to 1.2×10^{-14} (Theorem 6). The quadratic-hold bias Jacobian of Proposition 7 checks against finite differences to 1.7×10^{-13} .

Flight-estimator implementation validation: The FOH correction was implemented in the flight estimator itself (the driver-side composition routine `preintegrateImuStep`, which applies `exp_mixed` of Listing 1 once per IMU sample: per-sample two-point form with the $+\frac{T^3}{12}$ bracket in the code’s right-composition convention, sculling and scrolling included, bias Jacobians with the $\frac{T^2}{12}$ terms including the velocity-to-gyro-bias cross term) and exercised in a vehicle-model study distinct from this paper’s harness: IMU sub-sampled at 1 kHz into 100 Hz packets with a 200 Hz estimator (neither the deployed 1600 Hz rate nor the 5.76 kHz reference; case 7 below reruns one point at the reference rate to connect the decade), truth owned by the simulation at $\geq 20\times$ oversample, a coning pair plus phase-locked sculling plus a sub-Hz maneuver, 10 s integrations. The sensor path provides raw instantaneous rates, and the pre-FOH composition held the interval *end* sample (a backward rectangle), so the ZOH baseline carries a bounded $\frac{1}{2}\|\omega\| h$ phase-lag term on top of coning drift. Implementation verification: bit-for-bit mission identity with the feature flag off

```

if useFirstOrderHold then
  // right increment = T*N0 + (T^2/2)*N1 + (T^3/12)*[N0,N1]
  // [elided: subtract the bias anchor from both endpoint
  samples]
  angularVelocityDelta_rad_s :=
    angularVelocityMeasuredBodyFlu_rad_s
    - previousAngularVelocityMeasuredBodyFlu_rad_s;
  specificForceDelta_m_s2 :=
    specificForceMeasuredBodyFlu_m_s2
    - previousSpecificForceMeasuredBodyFlu_m_s2;
  rotationIncrement_rad := (startAngularVelocity_rad_s
    + 0.5 * angularVelocityDelta_rad_s) * dt
    + (dt * dt / 12.0)
    * LieGroups.S03.Quat.wedge(startAngularVelocity_rad_s)
    * angularVelocityDelta_rad_s;
  velocityIncrement_m_s := (startSpecificForce_m_s2
    + 0.5 * specificForceDelta_m_s2) * dt
    + (dt * dt / 12.0)
    * (LieGroups.S03.Quat.wedge(startAngularVelocity_rad_s
    )
    * specificForceDelta_m_s2
    - LieGroups.S03.Quat.wedge(
    angularVelocityDelta_rad_s)
    * startSpecificForce_m_s2);
  positionIncrement_m := -(dt * dt / 12.0) *
    specificForceDelta_m_s2;
else
  // [elided: zero-order hold, unchanged]
end if;
// [elided: bias Jacobians gain one (dt^2/12) cross term per
channel]
updatedExtendedPose := LieGroups.SE23.Quat.exp_mixed(
  previousExtendedPose,
  cat(1, positionIncrement_m, velocityIncrement_m_s,
    rotationIncrement_rad),
  zeros(9), [0.0, dt; 0.0, 0.0]);

```

Listing 2. The FOH per-sample increment construction as shipped (verbatim; unchanged

ZOH scaffolding and bias-Jacobian terms elided). The three assignments are exactly (16): trapezoid mean plus coning, sculling, and scrolling cross terms, formed from sample differences to avoid single-precision cancellation. Source: Estimation/StrapdownINS/preintegrateImuStep.mo, modelica_models branch foh-preintegration.

TABLE IV
FLIGHT-ESTIMATOR STUDY: ATTITUDE ERROR AT $t = 10$ s (RAD), ZOH (HELD END SAMPLE) VS FOH, AND THE PHASE-BLIND INTERPOLATION BOUND OVER THE MEASURED FOH ERROR.

case	f	ϕ	h [s]	ZOH	FOH	ZOH FOH	bound FOH
1	100	0.5	1e-3	4.47e-4	4.65e-5	9.62	3335
2	200	0.5	1e-3	6.80e-4	3.48e-4	1.95	3565
3	400	0.5	1e-3	2.36e-3	2.13e-3	1.11	4655
4	100	2.0	1e-3	1.00e-3	7.67e-4	1.31	808
5	200	2.0	1e-3	4.10e-3	5.66e-3	0.72	876
6	400	2.0	1e-3	2.99e-2	3.45e-2	0.87	1149
7	200	0.5	1.736e-4	9.06e-5	1.18e-5	7.71	3185

(29,810,344 cells, 0 mismatches); finite-difference verification of all five FOH bias Jacobians, including the position channels this paper’s harness left unexercised; a bracket-sign unit test (the wrong sign doubles the trapezoid error, as in Section III); and an OpenModelica cross-check at 4–5 significant digits. Listing 2 shows the shipped correction verbatim: the fifteen-line claim of Section VIII made real, the increments of (16) line for line.

Findings, with the honesty kept (Table IV; velocity and position columns behave like attitude, ratios 1.24–10.15 and 1.55–10.64 at 1 kHz, 19.7 and 20.5 at 5.76 kHz):

- 1) ZOH matches the coning prediction $\frac{1}{2}\phi^2(2\pi f)(2\pi fh)^2/12$ to 13–42% where coning dominates (the 2 mrad rows); the 0.5 mrad rows are dominated by the non-accumulating backward-rectangle phase term instead.
- 2) Measured FOH drift sits $808\times$ to $4655\times$ below the phase-blind interpolation bound, *independently reproducing* the harness tightness factors 389–4509 of Table XII: a second implementation cross-validates the bound’s looseness characterization (Remark 5).
- 3) Case 2 vs case 7 (same signal, h ratio 5.76): FOH error drops $29.6\times$ against the $(2\pi fh)^2 = 33$ theory scaling.
- 4) *Negative result, stated prominently (Fig. 5)*: at 1 kHz with $\phi = 2$ mrad and $f \geq 200$ Hz, FOH attitude error is $0.72\times$ to $0.87\times$ of ZOH, i.e. slightly worse: at 2.5 to 5 samples per vibration period the linear-interpolant model error is systematic and exceeds the bounded phase lag of the held-sample baseline. This confirms the bound structure (the interpolation leg dominates exactly where Theorem 4 says it must) and yields a practical admissibility condition: FOH pays uniformly for $2\pi fh \lesssim 0.7$ (at least nine samples per vibration period), and can lose to the held-sample baseline for $2\pi fh \gtrsim 1.2$ when coning dominates.
- 5) *Consequence at the deployed rate, stated with the same discipline*: the deployed configuration is $f_s = 1600$ Hz, and the motor-vibration band of interest sits at the silicon low-pass corner, $f \approx 200$ Hz, giving $2\pi fh \approx 0.79$. That is inside the marginal band of the condition just stated: above the 0.7 line below which FOH pays uniformly, and below the 1.2 line above which it can lose. *First-order hold is therefore not a clear win at the deployed configuration*. The evidence brackets it rather than settling it: at exactly $2\pi fh = 0.785$, Experiment 11 (Section VII) measures FOH at $0.137\times$ to $0.208\times$ of a held-interval-start ZOH, a clear gain, while the nearest measured points of this study, at $2\pi fh = 1.26$ against a held-interval-end baseline, put FOH at parity or slightly worse. Which ZOH variant and which vibration phase the deployed system presents decides the sign of a small effect. The constructive route is Section V: the centered quadratic hold converges one order faster in h and, measured at the deployed point, is a further $0.16\times$ to $0.21\times$ below FOH ($0.029\times$ to $0.034\times$ of ZOH). At 1600 Hz the centered hold, not plain FOH, is the recommended configuration, and the delayed-fusion architecture (Remark 6) is precisely what licenses it at zero added lag. At the 5.76 kHz reference rate ($2\pi fh = 0.22$ at 200 Hz) FOH is uniformly $7.7\times$ to $20\times$ better, which is what makes the crossover visible across the decade rather than a claim about the deployed vehicle.
- 6) Mission level: a 45 s GPS mission shows navigation RMSE 0.148753 m (ZOH) vs 0.148903 m (FOH), no measurable change, because the simulated plant carries no motor-frequency vibration and aiding bounds the

error. Stated plainly: the correction matters for the vibration regimes above, not for this benign aided mission.

Fig. 4 shows the drift trajectories themselves for three representative cases, against the endpoint aggregates of Table IV and Fig. 5. Mission-level and cross-stack validation of the estimator that hosts this correction is Part II’s Section XIX.

Reproducibility: `modelica_models` branch `foh-preintegration`, study `model` `Tests/PreintegrationHoldStudy.mo` with its scenario definitions; the feature is flag-gated and default-off in `preintegrateImuStep`. The drift series of Fig. 4 regenerate by running the study scenario `Tests/rumoca-scenario.preintegration-hold.toml` with `rumoca sim -c`, decimating the per-case attitude-error channels to a 20 ms grid (quote-aware CSV extraction; the full-rate output is large).

Reproducibility: All harness experiments run in one standalone Rust program, `foh-check` (about 1300 lines, no external crates): `xorshift64*` RNG with fixed seeds (42, 20260824, 7, 101–104, 555, 777, 999, 1234, 31415, 2718, 424242, 515151), RK4 truth at 10^4 substeps (or 24 per sample interval on the 1 s horizons) in `f64`, matrix exponential by scaling and squaring with a 24-term series, and the graded-logarithm identification described above. The harness emits the plotted data on each run (`data/sweep.dat`, `data/remainder.dat`, `data/tightness.dat`, tracked in the repository next to `main.tex`); Figs. 1, 2, and 3 are `pgfplots` renderings of those files, and Fig. 5 plots the flight-estimator numbers of Table IV.

[REVIEW] Decide where the harness will be published (e.g. the CogniPilot cyecca repository, as for [2]) and insert the link.

VIII. IMPLEMENTATION AT FLIGHT RATES

The forms below are written for FOH; they apply unchanged to the centered quadratic hold of Section V (three additional cross terms per sample), and the resulting deltas and Jacobians feed the reapplication path of Section VI as-is. As of this writing, FOH is implemented in `preintegrateImuStep` (`modelica_models` branch `foh-preintegration`), flag-gated and default-off, with the validation of Section VII; the identified next hold-error target is the retrodiction side, where measurement-age alignment currently uses a single held sample over ages up to 100 ms (`retrodict`, Section VI).

Recommended configuration at the deployed rate. At $f_s = 1600$ Hz with motor vibration at the 200 Hz filter corner, plain FOH is marginal ($2\pi fh \approx 0.79$, Section VII); the centered quadratic hold is the recommended configuration, measured a further $5\times$ to $6\times$ below FOH there (Table III). It is admissible only because fusion is delayed (Remark 6): the fused state may use a stencil that reaches one sample past each interval, while the live edge keeps the causal form. This is the clearest practical payoff of the delayed-fusion architecture in this paper.

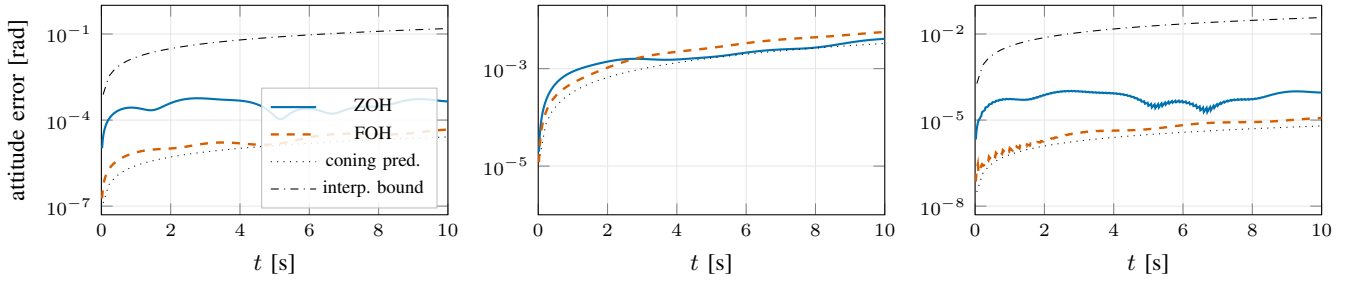


Fig. 4. Drift trajectories against oversampled truth, three representative cases of the flight-estimator study (`data/drift_case{1,5,7}.dat`, 20 ms decimation). Left: case 1 (100 Hz, 0.5 mrad, 1 kHz), FOH removes the hold error. Center: case 5 (200 Hz, 2 mrad, 1 kHz), five samples per vibration period: the linear-interpolant model error is systematic and FOH slightly exceeds ZOH, consistent with the crossover of Fig. 5; ZOH tracks its secular coning prediction (dotted) where coning dominates. Right: case 7 (200 Hz, 0.5 mrad, 5.76 kHz), FOH wins 7.7 $\times$. The 0.5 mrad cases ride the bounded backward-rectangle phase term on top of the coning line. The dash-dotted phase-blind interpolation bound (winning cases) sits orders above the measured FOH curves, as in Fig. 2.

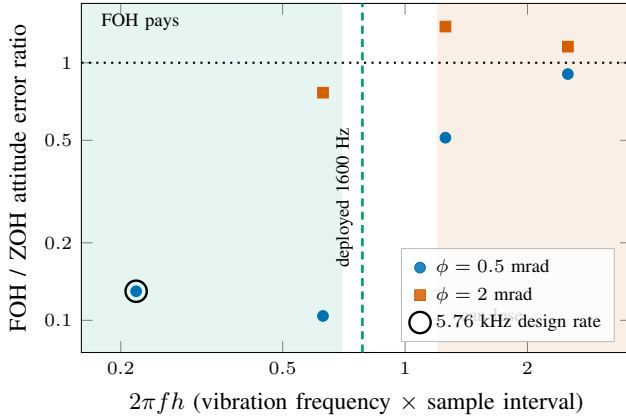


Fig. 5. The admissibility crossover, from the flight-estimator study (all seven cases of Table IV). Below the dotted line FOH beats the held-sample ZOH baseline. Practical rule: FOH pays uniformly for $2\pi fh \lesssim 0.7$, and can lose above ≈ 1.2 when coning dominates ($\phi = 2$ mrad); the circled point is the 5.76 kHz reference rate, where FOH is 7.7 $\times$ better on the same signal as the adjacent 1 kHz point. The deployed configuration (1600 Hz, vibration at the 200 Hz filter corner) sits at $2\pi fh \approx 0.79$, marked: inside the marginal band, so FOH is not a clear win there. The centered quadratic hold of Section V is the delayed-regime remedy, measured a further 5 $\times$ to 6 $\times$ below FOH at that point (Experiment 11).

[REVIEW] Architectural gap between the analyzed and the as-built system, stated plainly. Every hold result in this paper presupposes that IMU samples between estimator releases are *accumulated* into a delta packet. The flight firmware does not do this today: the estimator process consumes one subscription and returns early on the release check, the IMU-to-eFMU copy passes instantaneous angular rate and specific force rather than delta-angle and delta-velocity, and the driver decodes the latest FIFO entry, so samples between releases are discarded. The Modelica model matches the firmware in this respect, holding a single sample per interval. The composed sub-sample architecture is the *intended* design, as recorded by the `imuPreintegrationPeriod` parameter and by the coning-corrected delta fields of the `ImuSample` record, but it is not what either artifact implements at present. Accumulating every sample into a delta packet is therefore the precondition for any hold-order result here, FOH or centered, to apply on hardware; until then the deployed system is a one-sample-per-release ZOH

integrator and the results of Sections VII and VIII describe the architecture it is designed to become.

Per-sample form: Keep one closed-form update (Theorem 1) per IMU sample; each right increment is (16) built from two consecutive raw samples. This is FOH-exact through $\Xi_{\leq 3}$ between samples and costs three cross products and a handful of additions per sample beyond ZOH: one Lie bracket.

Per-step multi-sample form (preferred): One closed-form update per estimator tick whose right increment is the accumulated trapezoid deltas plus the summed cross terms over the interval's samples (eight at 1600 Hz with the 200 Hz reference tick), the group-derived analogue of Savage's multi-sample algorithms [8], [4]. The change to the existing preintegration routine is on the order of fifteen lines, and the scrolling term uses an input slot that is identically zero under ZOH.

[REVIEW] Open question (honest status): composing the per-sample corrections (16) reproduces the *polynomial-fit* class of multi-sample coning/sculling coefficients by construction, since both derive from piecewise-linear inputs at the same truncation order. It does *not* reproduce Savage's frequency-tuned coefficient sets, which are optimized against pure coning motion by design. The explicit coefficient-by-coefficient comparison for the five- and six-sample algorithms has not been carried out; at a sample-to-coning frequency ratio near 28 the difference is expected below the gyro noise floor, but this should be decided empirically in the validation harness.

Numerical conditioning: Form the coning term as $\omega_k \times \Delta\omega$ (not $\omega_k \times \omega_{k+1}$, which is algebraically equal but subtracts nearly parallel vectors) and the sculling term as $\omega_k \times \Delta a - \Delta\omega \times a_k$; the consecutive-sample forms are cancellation-prone in single-precision generated code. The FOH terms perturb the arguments of the closed-form coefficient functions (7) by $O(T^2 \|\omega \times \Delta\omega\|)$ and add only polynomial terms, so they are additive corrections independent of the small-angle branch selection in (7); at these rates the rotation argument is at least two orders inside the series branch.

Error budget: Table V gives representative numbers for the vibration case of Section I ($\phi = 0.5$ mrad on two axes, $f = 200$ Hz, true coning rate about $0.009^\circ/\text{s}$; residual scale factor for an uncorrected integrator approximately $(2\pi f/f_s)^2/12$).

TABLE V

CONING ERROR BUDGET (REPRESENTATIVE; $f = 200$ HZ VIBRATION AT $\phi = 2$ MRAD, TRUE CONING RATE $\approx 0.009^\circ/\text{s}$, GYRO BIAS-INSTABILITY FLOOR $\approx 0.0092^\circ/\text{s}$). HOLD ROWS SCALE THE COMPOSED FACTOR BY THE MEASURED EXPERIMENT 11 RATIOS AT THE DEPLOYED POINT.

scheme	residual factor	residual drift	vs. floor
ZOH, single 1 kHz interval	~ 0.13	$\sim 1.2\text{e-}3^\circ/\text{s}$	marginal
composed at 1600 Hz (deployed)	$\sim 5.1\text{e-}2$	$\sim 4.6\text{e-}4^\circ/\text{s}$	$20\times$ below
composed at 5.76 kHz (ref.)	$\sim 4.0\text{e-}3$	$\sim 3.6\text{e-}5^\circ/\text{s}$	$250\times$ below
+ FOH (16) at 1600 Hz	$\sim 1.1\text{e-}2$	$\sim 9.6\text{e-}5^\circ/\text{s}$	$95\times$ below
+ centered quad. at 1600 Hz	$\sim 1.8\text{e-}3$	$\sim 1.6\text{e-}5^\circ/\text{s}$	$580\times$ below

[REVIEW] The budget numbers in Table V rest on the standard coning scaling (the factor of 12 and the quadratic rate ratio); the composed rows are evaluated at $(2\pi f/f_s)^2/12$ for $f = 200$ Hz, giving 5.1×10^{-2} at the deployed $f_s = 1600$ Hz and 4.0×10^{-3} at the 5.76 kHz reference. The hold rows multiply the deployed composed factor by the measured Experiment 11 ratios (0.208 for FOH, 0.034 for the centered quadratic, at $f = 200$ Hz and $\phi = 2$ mrad); those ratios are measured against a held-interval-start ZOH, while the flight-estimator study of Section VII uses a held-interval-end baseline, so the two are not interchangeable row for row. The table has not been re-derived independently and should be regenerated by the simulation harness before publication.

Part II: The RDD2 Delayed-Fusion Error-State Filter

Part II documents the filter that Part I’s propagation engine serves: the navigation estimator of the RDD2 multirotor, a 15-state error-state Kalman filter (ESKF) on $SE_2(3) \times \mathbb{R}^6$ with delayed aiding fusion, written entirely in Modelica, compiled to single-precision embedded C for a Cortex-M7 flight target, and validated in closed-loop simulation and standalone replay against PX4 EKF2 and ArduPilot EKF3 on identical sensor streams. Part II never re-derives Part I’s results; it states interfaces and consumes theorems. The implementation (modelica_models branch foh-preintegration) is Part II’s ground truth: every statement labeled *audited* is read from the named source file.

IX. THE FILTER: STATE, ERROR, AND SYMMETRY SCOPE

A. Nominal state and error state

The nominal state is $X = (p, v, q, b_g, b_a)$: world position (ENU, m), world velocity (m/s), attitude quaternion (world from body, FLU body axes), and gyroscope and accelerometer biases (body frame). The pose triple $(R(q), v, p)$ is the $SE_2(3)$ element of Part I, carried as the implementation’s 10-vector; the biases are additive states in $\mathbb{R}^6$ (audited: ESKF/State.mo, NominalState.mo). The pose propagates by exactly Part I’s mixed flow $\dot{X} = MX + XN$ on the time-extended algebra, i.e. on $\text{Gal}(3)$ (Section VI-C).

The error state is the 15-vector $\varepsilon = \{\varepsilon_p, \varepsilon_v, \varepsilon_\theta, \varepsilon_{b_g}, \varepsilon_{b_a}\}$ with right injection on the pose and additive error on the biases.

Lemma 6 (error-convention reconciliation across Parts). *Part II defines the error truth-relative-to-estimate,*

$$X_{\text{true}} = \hat{X} e^{[\varepsilon_{\text{pose}}]^\wedge}, \quad b_{\text{true}} = \hat{b} + \varepsilon_b, \quad (41)$$

so that the derived measurement-Jacobian signs match the code. This is the same convention as Part I: Part I’s conventions audit (Section VI) established that injection applies the correction on the right, $\hat{X} \mapsto \hat{X} e^{[\delta]^\wedge}$ (audited: ESKF/inject.mo), and the injected correction $\delta = Kr$ is an estimate of ε in (41): applying it moves the estimate toward the truth on the same side. No sign flip or convention change is needed anywhere between Parts; the transition, reset, and recorection routines all sit on this same right side (the conventions audit found no disagreement).

In the taxonomy of the equivariant-symmetries literature [12], this filter is the *left imperfect-IEKF* geometry: symmetry group $SE_2(3) \times \mathbb{R}^6$, direct product on the bias, with an error transition that depends only on the IMU inputs and never on the state (the A matrix of Section XIII). Reported taxonomy plus proved identification (companion equivariance analysis).

B. Why not the bias-inclusive tangent group

The companion analysis examined adopting the tangent-group symmetries ($SE_2(3) \ltimes \mathfrak{se}_2(3)$, and the Galilean form $\text{Gal}(3) \ltimes \text{gal}(3)$ of [11]) and concluded: no benefit for this vehicle under current requirements. Four grounds, with statuses: (i) the mean propagation is already the Galilean machinery and the closed forms lift verbatim with zero additional truncation (Remark 8: fiber slaving; proved and verified numerically), so there is nothing to adopt on the integrator side; (ii) exact bias transport is impossible in any framework (path dependence, Section VI-C), so the tangent group cannot remove the $O(\|db\|^2)$ remainder either; (iii) the demonstrated tangent-group payoffs are covariance consistency in regimes RDD2 does not enter (0.5 to 1.0 s open-loop windows against RDD2’s 0.2 s horizon, where the reported EuRoC NEES gain is 1 to 2 percent median; 20-degree initial attitude errors against pad alignment), and the left imperfect-IEKF column already matches the tangent-group filter’s asymptotic ANEES in the literature’s own Monte Carlo (reported); (iv) cost moves the wrong way on the binding constraint: 18 error states push the measured 1.59 ms worst-case step bound to an estimated 2.3 to 2.75 ms, 20 states to 2.9 to 3.8 ms, and the 1 kHz ambition dies there (estimate grade on measured baselines). A fully specified prototype gate (two covariance arms sharing all mean machinery, a scripted 1 to 2 s outage plus a 20-degree misalignment variant, NEES calibration as the decision metric, one to two days on existing rigs) converts this decision the day a requirement appears for calibrated covariance through outages longer than about half a second (Section XX).

X. PROPAGATION INTERFACES

A. The driver-side preintegrator

`preintegrateImuStep` (Listing 2) composes raw IMU samples at the IMU rate into a *preintegration packet*: per sample one `exp_mixed` application (Proposition 1) with the FOH correction of Theorem 1 behind the `useFirstOrderHold` flag, accumulating simultaneously the five bias Jacobians of the packet including the FOH $T^2/12$ cross terms (Proposition 4). In the reference vehicle wiring (audited: `Vehicles/Rdd2/WaypointVehicleSystem.mo`) the IMU samples at 1 ms, packets close every 10 ms, and the estimator ticks at 5 ms, so the estimator sees a fresh packet every other tick; the flight-codegen study separately exercises the block at 1 kHz (the rates are parameters, not structure). Each packet carries the timestamp and integration time, the delta triple, the bias linearization anchor, and the five Jacobian matrices; the estimator’s finiteness scan checks every field before ingestion (audited: `ESKF/step.mo`).

B. The bias anchor and packet recorection

`correctPreintegratedImu` moves a stored packet from its anchor to the current bias estimate at first order, $\Delta q \mapsto \Delta q \otimes e^{[J_{Rdb}]^\wedge}$ with additive Jdb on the velocity and position deltas: exactly the recorection whose remainder Proposition 8 bounds (measured slope 1.992, constant 0.063 of the bound). The anchor’s causality is explicit in the wiring: each packet linearizes the *next* window around the bias the estimator published on its *previous* step, through a `pre()` boundary. That boundary closes a same-instant discrete loop that the compiler’s same-tick refusal caught during development; the model fix inserted three `pre()` boundaries and moved the mission RMSE from 0.1490 to 0.1488 m, within noise (audited plus measured).

C. What predict needs from Part I

`predictPreintegrated` consumes one packet and performs the nominal update in exactly the reapplication form (37) and the covariance update $P^+ = \Phi P \Phi^T + Q_d$ with the packet transition and Simpson noise integral of Section XIII. The interface contract, stated once: *predict* needs from the preintegrator the delta triple, the anchor and five Jacobians, and the integration time; and it needs from Part I the guarantees that the delta is the exact ZOH (optionally FOH) flow (Proposition 1, Theorem 1), that reapplying it on a corrected initial state is exact (Theorem 6), and that moving it to a new bias has a second-order remainder (Proposition 8).

XI. DELAYED FUSION AS IMPLEMENTED

A. The transported-Jacobian architecture

The implemented delayed fusion is retrodiction-based, the measurement-age-alignment variant of Part I’s buffered-horizon picture. The filter state lives at the live edge; each aiding sample carries a capture timestamp and its age is computed against the held IMU time; the correction function projects the *current* state back to capture time with `retrodict`,

evaluates the residual and the measurement Jacobian H_d *there*, transports the Jacobian to the current tangent by the error-flow factor

$$H = H_d \Phi(-\text{age}), \quad \Phi(-\text{age}) \approx e^{-A \text{age}} \quad (42)$$

(third-order Taylor polynomial, evaluated at the current held IMU sample), and applies the correction to the current state through the common linear path of Section XII. Admission: future-stamped samples and ages beyond `maximumAidingDelay_s` (default 0.25 s) reject with a named cause. In the reference wiring, GPS latency 0.1 s plus up to one 0.1 s sample period gives ages near 0.2 s (Part I’s “approximately 200 ms”); optical-flow ages are 10 to 20 ms. Audited: every `correct*` function carries the identical stanza.

No measurement buffer is replayed: the state-independence of the mixed-flow factors (Theorem 6) makes retrodiction a reapplication of *inverse* factors, $X_d = e^{-M \text{age}} X e^{-N \text{age}}$ with N built from the single current bias-corrected IMU sample held over the age (audited: `ESKF/retrodict.mo`, quoted in the review box of Section V). The gravity and coupling terms are exact for any age; the single-held-sample body factor is the hold approximation Part I names as the next hold-error target (Section XX).

B. Relation to the buffered-window form

Part I’s delayed-fusion section and reapplication theorem are written for the buffered-window architecture: *correct at the horizon*, then reapply the buffered preintegrals exactly. The implementation realizes the equivalent transported-Jacobian form: *correct the current state with H transported from the horizon*. The two are first-order equivalent, sharing the same mean-transport exactness theorem; the transported form trades buffer memory and a reapplication pass for the single-held-sample retrodiction approximation. The centered-stencil admissibility of Remark 6 applies to the buffered form and is not exercised by the current implementation: it remains the design headroom the architecture was chosen to admit. An explicit first-order equivalence lemma between the two forms is an open item (Section XX).

Verification status of the pair $(H_d, \Phi(-\text{age}))$: a mechanized-proof campaign synthesized the measurement Jacobians from the Modelica measurement models for barometer, GPS position, and magnetometer yaw and matched the shipped hand-derived H_d to the floating-point floor (worst 1.33×10^{-15} ; central finite differences to 7.1×10^{-9} over a randomized sweep; artifact: `modelica_models` branch `lie-jacobian-eskf-proof`, commit 17767d3). Caveat, carried verbatim: only H_d is proven; the full $H = H_d \Phi(-\text{age})$ including the delay transport was not synthesized, so H is mechanically verified exactly only at zero aiding delay; the transport factor rests on the block derivation above and on mission-level validation (open item, Section XX).

C. Source supervision: anchor, staleness, recovery ladder

ESKF/step.mo supervises fusion; its source comments carry measured failure narratives for every design decision, and this subsection is deliberately prose-dense: the supervision layer is where the measured failure modes live. Audited throughout; the embedded numbers are measured per the source narratives.

- Cross-clock delivery is level-triggered: producers hold the latest packet; timestamp novelty makes consumption exactly-once. Staleness clocks run per source in wall-clock seconds, cleared on a fresh usable sample; driver valid duty cycles are explicitly not trusted (the deployed GPS driver pulses valid).
- The *anchor* is the most authoritative source still delivering: mocap, then GPS, then optical flow (barometer and magnetometer are never anchors); it is latched between samples. A rejection clock runs on the anchor alone: advanced while the anchor has not accepted, cleared by anchor acceptance or a genuine anchor change, and *frozen* (not cleared) on total aiding loss.
- Recovery ladder on that clock: stage 1 (default 1.0 s) begins *ramping* position and velocity covariance toward the mission envelope (Section XIII), leaving the state untouched, with anchor exclusivity during the widened-gate window so the fastest sensor cannot steal re-acquisition (measured counterexample: optical flow recapturing the filter at -179 m/s against 7 m/s truth without exclusivity); stage 2 reports `RecoveryAidingDivergent` and does nothing else: it does not adopt, re-seed, or withdraw `estimate.valid` (on the deployed stack `estimate.valid=false` zeroes the motors and latches; the source comment records the witnessed crash mechanism). Misconfigured windows disable the ladder affirmatively.
- `estimate.valid` answers exactly one question: are the published numbers finite and initialized. IMU-derived outputs hold the last finite sample rather than publishing NaN or zero (a zero body rate would assert “not rotating” to the rate loop, Remark 7).
- At most one aiding correction applies per tick, in authority order: mocap; GPS (joint, then position-only, then velocity-only); barometer; optical flow; magnetometer. Initialization and commanded reset seed position and attitude from the most authoritative available source (mocap, GPS, TRIAD-style accelerometer-plus-magnetometer alignment, declared initial state), and every seed must prove finiteness first.

XII. MEASUREMENT MODELS AND CORRECTIONS

All statements audited against the named sources; the H_d rows for barometer, GPS position, and magnetometer yaw are additionally mechanically verified (Section XI). Conventions: under (41), to first order $p_{\text{true}} = p + R\varepsilon_p$, $v_{\text{true}} = v + R\varepsilon_v$, $R_{\text{true}} = R\text{Exp}(\varepsilon_\theta)$, so a world-frame quantity differentiated in the local tangent picks up one factor of R ; the implementation removes it by expressing every innovation *in the body*

frame of the delayed attitude estimate, which turns the position and velocity Jacobian blocks into constant identity blocks (the hallmark of the invariant-EKF measurement structure), with the measurement noise rotated into the same frame ($R_d^T \text{Cov } R_d$): information re-expressed, not moved. Neglected term, stated once for all body-frame residuals: rotating the innovation by the estimated rather than true delayed attitude contributes $\varepsilon_\theta \times r_0$, second order in small quantities (standard ESKF practice; audited absent from the code; a one-paragraph bound tying it under the gate threshold is an open item).

A. GPS position and velocity

Model $z_p = p + n_p$, $z_v = v + n_v$ in world ENU with sensor-reported covariances. Residual at the delayed state, $r = \{R_d^T(z_p - p_d), R_d^T(z_v - v_d)\}$. Since $z_p = p_d + R_d\varepsilon_p + n_p$, $R_d^T(z_p - p_d) = \varepsilon_p + R_d^T n_p$ and identically for velocity:

$$H_d = \begin{bmatrix} I_3 & 0 & 0 & 0 & 0 \\ 0 & I_3 & 0 & 0 & 0 \end{bmatrix} \quad (43)$$

exactly, with zero attitude coupling; noise blocks $R_d^T \text{Cov } R_d$, position-velocity cross-covariance not representable in the packet and assumed zero. The joint 6-dof form runs when both validity flags are set; 3-dof forms otherwise. Audited (`correctGps*.mo`); the position block mechanically verified to 1.33×10^{-15} .

B. Optical flow with co-timed range

The packet carries integrated line-of-sight angles, co-integrated gyro angles, a co-timed ground distance d along the nadir ray, the integration time Δt_f , per-channel covariances, and a quality scalar. Velocity manufacture (audited, including the sign-convention comment): gyro compensation then range scaling, $v_{b,1}^{\text{meas}} = d\phi_2^c/\Delta t_f$, $v_{b,2}^{\text{meas}} = -d\phi_1^c/\Delta t_f$ with $\phi^c = \text{flow} + \text{gyro}_{1,2}$, d floored at the near-field guard and Δt_f at 1 ns. Terrain is deliberately *not* used in the manufacture: the range is the sensor’s own co-timed measurement, and terrain remains a separately filtered auxiliary state. Measurement model $h(X) = (R^T v)_{1:2}$; residual against the delayed state. With $R_{\text{true}}^T v_{\text{true}} = R^T v + \varepsilon_v + [R^T v]^\wedge \varepsilon_\theta + O(\varepsilon^2)$,

$$\begin{aligned} H_d[1:2, :] &: \text{velocity block } [I_2 \mid 0], \\ &\text{attitude block } [R_d^T v_d]^\wedge[1:2, :], \end{aligned} \quad (44)$$

exactly as coded: the attitude coupling is genuine here (a tilt error rotates the projected velocity), unlike GPS. The measurement covariance propagates the manufacture exactly: the index-swap and sign pattern of the linear flow-to-velocity map applied to the summed line-of-sight and gyro covariances, scaled by $(d/\Delta t_f)^2$, plus the rank-one range-uncertainty term with the analytic velocity-to-range derivative. Admission beyond the common path: all fields finite, channel variances positive, $\Delta t_f > 0$, d above the near-field guard (default 0.2 m), quality at least the threshold (default 0.2). The simulated producer intersects a 9-feature ray grid with the ground plane using exact projective rate equations and preserves plane-inclination effects, so the estimator’s planar-ground assumption is genuinely exercised rather than baked into the producer (audited: `simulateOpticalFlowPlane.mo`).

C. Magnetometer yaw

Manufacture (audited: magnetometerYaw Observation.mo): tilt-compensate the raw body field with the leveling rotation built from the delayed pitch and roll only; the measured heading is the wrapped difference of the declination (from the world reference field's horizontal direction only; no declination grid) and the leveled-field azimuth; the heading variance propagates through the atan2 gradient. Residual: wrapped scalar heading difference against the B321 yaw of the delayed quaternion. The Jacobian row is

$$H_d[1, 7:9] = [0, \sin \phi / \cos \vartheta, \cos \phi / \cos \vartheta], \quad (45)$$

precisely the third row of the body-rate-to-Euler-B321-rate kinematic matrix: yaw responds to body-frame attitude error about body y and z through the tilt geometry, and magnetometer data can never directly move roll or pitch. Guard $|\cos \vartheta| > 0.1$, else rejection rather than division by a vanishing cosine. Mechanically verified as part of the H_d proof set.

D. Barometer with learned pressure datum

Model $z = p_z + b_{\text{baro}} + n$. Residual $r = z - b_{\text{baro}} - p_{d,3}$; Jacobian row $H_d[1, 1:3] = R_d[3, :]$ (third row of the delayed DCM, from $p_{\text{true}} = p + R\varepsilon_p$); variance $\text{Var}(z) + \max(\text{Var}(b_{\text{baro}}), 0)$. Mechanically verified. Datum handling, two mechanisms (audited): a stationary startup window (default 25 samples) learns the datum by a scalar Kalman recursion while the barometer is *withheld* from navigation correction, removing the positive feedback of one observation updating both altitude and its own datum; and after every accepted barometer correction, a rank-one *posterior floor* adds back $\max(\text{Var}(b_{\text{baro}}) - \sigma_{\text{vert}}^2, 0)$ along the local vertical $R_d[3, :]$: the learned datum is one common nuisance variable across packets, not fresh independent noise, so the filter must not converge its vertical below the datum uncertainty. The floor keeps the posterior honest without a sixteenth state.

E. Motion capture

Position plus attitude, 6-dof, fused at current time (wired without a transport-latency model; adding one is a wiring change, not a filter change). Residuals $r_p = R^T(z_p - p)$ and $r_\theta = \text{Log}(q^{-1} \otimes q_{\text{meas}})$, the right attitude error; both Jacobian blocks are exact identities under (41) (the attitude block's exact $J_r^{-1}(r_\theta)$ is approximated by identity, second order in the residual). Audited: correctMocap.mo.

F. Auxiliary scalar states

Terrain altitude runs as a scalar Kalman filter on locally horizontal terrain (observation $z_t = p_z - d R_{33}$, guarded by a tilt cosine floor of 0.5 and quality), published as an auxiliary estimate that does *not* feed back into the navigation correction. A full ray-to-plane range observation row (opticalFlowRangeJacobian.mo) exists and is test-covered but is *not* wired into the correction chain: fusing range directly into the navigation state is a designed-but-unshipped option, recorded here so this paper does not imply otherwise.

G. The common correction path: gating, trust region, Joseph

Every measurement takes one path (audited: correctLinear.mo, solveSPD.mo, josephUpdate.mo, conjugateReset.mo); $m \in \{1, 2, 3, 6\}$.

- 1) *Affirmative admission*: every residual and covariance entry finite, every R diagonal strictly positive (a non-positive R can leave S factorizable while driving the gain above unity; both witnessed failure modes are recorded in the source). The scans are $O(m^2)$ against the $O(n^3)$ solve, and the factorization stays unconditional so worst-case timing is unchanged.
- 2) *Innovation machinery*: $C = PH^T$, $S = \text{sym}(HC + R)$; one Cholesky factorization serves both the gain solve and the gate, solving the augmented right-hand side $[C^T | r]$ in one call and returning K^T and the whitened residual together.
- 3) *NIS gate*: $\text{nis} = r^T S^{-1} r$, accepted iff finite, nonnegative, and at most 6.0 m (about 0.25% false rejection for $m = 2, 3, 6$). Acceptance is affirmative: a NaN anywhere fails *into* rejection (the source records the NaN-acceptance defect this replaced). Every rejection carries a named cause, ordered by depth: NotFinite, CovarianceUnusable, Factorization, Gate.
- 4) *Trust region on the rotation rows of the gain only*: if the attitude part of Kr exceeds the cap, gain rows 7:9 are scaled and the correction recomputed; position, velocity, and biases apply in full. Scaling the *gain* rather than the correction vector is load-bearing: the Joseph form is valid for an arbitrary gain, so state and covariance must use the same effective gain. The source records the measured failure of whole-vector scaling (a rate-dependent ratchet: monotone attitude walk to 1.43 rad with unbounded velocity divergence at 1 kHz aiding, against +0.062 m tracking for the row-scaled form).
- 5) *State update by right injection; covariance update* in Joseph form $P^+ = (I - KH)P(I - KH)^T + KRK^T$ (exact for any gain, including the trust-scaled one), symmetrized; *reset transport* by $\text{diag}(J, I_6)$ with J the $SE_2(3)$ right Jacobian of the applied correction (the reset whose convention consistency the Part I audit established).

solveSPD measures its pivot threshold at *runtime* against the executing precision (halving until $1 + \epsilon/2 = 1$): the same Modelica source runs as binary64 in simulation and binary32 in the generated flight code, and a fixed 10^{-12} tolerance, unreachable in binary32, previously degenerated the check. It returns `ok=false` rather than garbage on a non-positive-definite S , with the whitened residual zeroed so every downstream gate term stays well defined.

XIII. COVARIANCE MACHINERY

Tangent ordering $\{p, v, \theta, b_g, b_a\}$; twelve noise channels. All audited against the named sources.

A. Continuous error dynamics and noise input

With w, f the bias-corrected body rate and specific force (audited: `continuousTransition.mo`),

$$A = \begin{bmatrix} -[w]^\wedge & I_3 & 0 & 0 & 0 \\ 0 & -[w]^\wedge & -[f]^\wedge & 0 & -I_3 \\ 0 & 0 & -[w]^\wedge & -I_3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (46)$$

the body-frame (right-error) linearization of the mixed flow: input-dependent, state-independent, the defining property of the left imperfect-IEKF geometry. Gravity does not appear: it lives in the left factor and is handled exactly by the closed form, the mixed-invariant dividend at the covariance level. The noise input G routes gyro noise to attitude ($-I_3$), accel noise to velocity ($-I_3$), and the two bias random walks to their rows (I_3); the continuous Q_c is block diagonal from the tuning record with units in the names (defaults 10^{-5} , 10^{-3} , 10^{-8} , 10^{-6} times I_3).

B. Discretization and the packet transition

`discreteTransition` is the third-order Taylor polynomial $\Phi(dt) = I + Adt + (Adt)^2/2 + (Adt)^3/6$, dimension-generic (the same function serves the 15×15 transition and the negated-age delay transport (42)). `discreteProcessCovariance` is the Simpson quadrature of the held-input integral $Q_d = \int_0^{dt} \Phi(s) G Q_c G^\top \Phi(s)^\top ds$ on nodes $\{0, dt/2, dt\}$ (Section VI-D(iv)): exact for integrands of polynomial degree three, so with the third-order Φ the quadrature error is $O(dt^5)$ relative terms against leading $O(dt)$ mass, far below tuning uncertainty at 1 to 10 ms; congruences of PSD matrices with positive weights keep Q_d PSD by construction.

For a preintegrated packet, `predictPreintegrated` builds the *exact* local transition from the packet itself ($\Delta R, \Delta v, \Delta p, dt$, five Jacobians):

$$\Phi = \begin{bmatrix} \Delta R^\top & \Delta R^\top dt & -\Delta R^\top [\Delta p]^\wedge & \Delta R^\top J_{pg} & \Delta R^\top J_{pa} \\ 0 & \Delta R^\top & -\Delta R^\top [\Delta v]^\wedge & \Delta R^\top J_{vg} & \Delta R^\top J_{va} \\ 0 & 0 & \Delta R^\top & J_{Rg} & 0 \\ 0 & 0 & 0 & I_3 & 0 \\ 0 & 0 & 0 & 0 & I_3 \end{bmatrix}, \quad (47)$$

the adjoint-type transport of the right-tangent error through the delta (Section VI-D(i)) with the packet's bias Jacobians in the bias columns. Packet process noise reuses the Simpson integral with *equivalent held inputs* recovered from the delta itself, $w_{eq} = \text{Log}(\Delta q)/dt$ and $f_{eq} = J_l^{-1}(\text{Log}(\Delta q)) \Delta v/dt$: the constant rates that would reproduce the packet's increments.

C. Dropout, envelope, and reset discipline

IMU dropout (`holdCovariance`): on a tick with no usable IMU sample the nominal state holds but the covariance still steps through the *full* transition at zero corrected rates plus the noise integral. The source records why noise-only was wrong exactly where it matters: the dominant blind-interval position growth is $P_{vv} dt^2$ through the kinematic I_3 block,

which holds whatever the IMU does; omitting the transition left the position variance growing by exactly 0.0 in binary32 over a measured 10 s blind window. A third branch (held packet mid-accumulation) holds both state and covariance: propagating or adding noise twice would double-count.

Envelope bounding (`limitCovariance`): diagonals exceeding the mission envelope are pulled back by the two-sided diagonal congruence DPD , $D_i = \min(1, \sqrt{b_i/P_{ii}})$: symmetric, PSD-preserving, correlation-preserving, unlike clamping; envelope values sit where the state is already maximally uninformative, so capping loses no information while keeping the matrix inside binary32 Cholesky range.

Recovery inflation (`inflateCovariance`, ladder stage 1 only): position and velocity diagonals ramp toward the envelope by an additive increment. Three load-bearing decisions, each with a measured counterexample in the source: a ramp, not a jump (a jump saturates the gain and adopts whatever arrives first: +52 m position error, a lying flow sensor adopted at 0.72 m/s discrepancy instead of 3.63); position and velocity only (inflating attitude lets the filter explain residuals as rotation with nothing to correct it: $-35/+67$ m/s against 6 m/s truth); additive diagonal, not congruence (DPD preserves the innovation block's ill-conditioning exactly: NIS 76.8 against threshold 36 with Cholesky succeeding).

Reset conjugation (`conjugateReset`, Section VI): the sparse blockwise form collapses 54,000 multiplies to 2,430 structurally nonzero ones; deleted terms are exact zeros in trailing positions so no partial sum shifts; verified bit-identical through the generated C over a 4000-tick mission trace at both optimization levels, and exactly equal in an OpenModelica harness over 400 random covariances spanning nine decades. Measured payoff: aided tick $194.1 \rightarrow 99.9 \mu s$ at -02 . The same exact-zero discipline in `limitCovariance` measures $-30.7 \mu s$ per tick.

What Part I contributes here, consumed by reference: the headline bound and certificate bound the mean propagation error per interval; the remainder bound controls bias recorection; the reapplication theorem makes the delayed mean transport exact so no covariance is spent on retrodiction structure. Part II adds bookkeeping fidelity (Joseph for arbitrary gains, PSD-preserving caps, exact-zero sparsity), not new bounds; the NEES calibration of this pipeline is measured in Section XIX-C: translational channels calibrated, attitude optimistic by about $2\times$ during aggressive descent, a tuning-level finding.

D. Single-precision conditioning, measured

The flight artifact runs pure binary32; would a square-root or UD factorization be warranted? The question was settled by a frozen-criteria study (criteria H1 to H4 written down before the first run, so the data could not steer the bar; all numbers measured): the f32 artifact against a mechanical f64 twin of the *same* emitted source (three rewrite rules, nothing else, including evaluation order), bit-identical inputs, 3600 s at 1 kHz with 80 designed 5 s GPS outages as the classic conditioning stressor.

Verdict per criterion. H1, positive definiteness: the minimum eigenvalue of P stayed strictly positive at all 36,000 samples, settling at 3.16×10^{-7} (the converged gyro-bias variance) and matching the f64 reference to 4 significant figures at every sample. H2, symmetry: exactly zero asymmetry at every tick (the emitted symmetrize writes both triangles from one expression). H3, divergence: worst ratio 0.028 of the 0.1-sigma bar for 59.2 of 60 minutes; the f32-vs-f64 state difference is a flat, growthless rounding floor (position at most 8.4×10^{-3} m, attitude 1.2×10^{-4} rad). H4, interventions: zero `limitCovariance` clamps in the hour (diagonal peak 13.3 against a 10^4 envelope), recovery active 0.28% of ticks (entirely the designed outages), rejections 0.81%, and every counter bit-identical between the two precisions.

Mechanism. The filter is contractive: a single 1-ulp f64 perturbation of one GPS measurement decays to exactly zero state and decision difference by 600 s and stays zero for the remaining 50 minutes, so f32 rounding is a bounded equilibrium floor, not chaos. The raw condition number of P (baseline 10^5 to 1.2×10^6 , spiking to 4.3×10^7 at outage peaks, identical in f64) is almost entirely SI-unit disparity: the correlation matrix has condition median 12 and maximum 201, so $\epsilon_{32} \cdot \text{cond}(C) = 2.4 \times 10^{-5}$, five orders of margin. The cheap mitigation was tested per protocol and rejected: f64 kernel accumulation changes no decision (including the one flip below) at $1.61 \times$ kernel cost.

The one honest transient. At $t = 3553$ s a marginal magnetometer sample lands on opposite sides of the innovation gate in the two runs: 13 of 72,000 magnetometer decisions differ, in one burst decaying within 47 s; truth-referenced accuracy is identical to 5 significant digits and P agreement stays within 2.8% through the episode. By the frozen letter of H3 this is a degraded window (9 of 3600 checkpoints); the measured mechanism is a gate flip on a measurement within the rounding floor’s width of the threshold, a marginal-measurement phenomenon, not a covariance one. Gate hysteresis would remove the sensitivity and is noted as optional model-level hygiene, not a numerical necessity.

Conclusion. The covariance form with its existing hygiene (Joseph for arbitrary gains, explicit symmetrize, PSD-preserving envelope, pivot-guarded solve) is retained; UD or QR square-root is *not warranted* for this filter at this precision, on the frozen criteria; the cost lever is orthogonal to the factorization question (the dense-form contraction rewrites, since delivered). This subsection is the affirmative single-precision health record for the covariance path.

XIV. ANGULAR VELOCITY AND CONTROL INTERFACES

Brief by design; the scoped statement is Remark 7. The rate loop consumes the *estimator’s* published body rate, not the raw gyro: `navigationEstimate` publishes the held-finite bias-corrected sample, its world-frame rotation, and the corrected world acceleration $R(f - \hat{b}_a) + g$ (audited). This wiring is why `estimate.valid` is safety-critical (Section XI-C) and why publication holds the last finite IMU sample rather than zeroing. The controller consumes the estimate published on the

TABLE VI
MEASURED FLIGHT ARTIFACTS (FLASH TEXT, RAM INSTANCE).

artifact	flash [B]	RAM [B]
NavigationEstimator	58,936	31,980
GuidanceController	9,050	small
WaypointTrajectoryPlanner	31,806	n/m

estimator’s *previous* step through an explicit `pre()` boundary on the navigation bus, stating the causal order instead of leaving a same-instant loop for the scheduler (the compiler-refusal episode of Section X). The control interface never sees retrodicted states: retrodiction exists only inside the correction functions, and the live edge keeps the causal hold forms of Remark 6.

XV. THE UKF VARIANT

[REVIEW] This section is deliberately an *outline*: the UKF sources are audited at interface level, but the section needs one prose pass plus one table (about half a day), and the sources contain no head-to-head ESKF-vs-UKF benchmark number; that row is an open item (Section XX). It is included as an outline rather than padded into false prose.

Same `PartialEstimator` boundary, sensor set, and nominal-state representation as the ESKF. What differs (audited: `Estimation/StrapdownINS/UKF/`): sigma-point propagation replaces the linearized covariance step (31 points for $n = 15$; $\alpha = 1$, $\beta = 2$, $\kappa = 0$); sigma points are generated in the local tangent by Cholesky of P , injected on the manifold by the same right injection, propagated through the *same* closed-form preintegration packet with the delta moved through the stored first-order bias Jacobians per hypothesis, and re-centered by the local error log; process noise adds the same Simpson Q_d ; and corrections use statistical linearization with the same augmented SPD solve, NIS gate semantics, and attitude trust region. The Part I convention audit of this path (review box, Section VI) found no disagreement, and the linearized sigma-point bias coupling is consistent with Proposition 8.

XVI. IMPLEMENTATION AND COST

All numbers measured (flight-codegen record; method definitions in the embedded-efficiency dossier). Target: NXP MR-VMU-TROPIC (i.MX RT1064, Cortex-M7 at 600 MHz), `arm-none-eabi-gcc 15.2`, `-Os`, single precision throughout.

All three artifacts together are a small fraction of the 512 KB TCM (Table VI). Worst-case estimator step (GPS joint correction branch, model-A upper bound): 954,168 instructions, 82,118 MACs, 113,718 floating-point operations, $3.46 \times$ better than the pre-optimization baseline; at 600 MHz and one instruction per cycle this bounds the step at about 1.59 ms (the M7 dual-issues, so the true figure is lower). The honest statement, carried verbatim from the record: fits with margin at 250 to 500 Hz today; 1 kHz needs the remaining hotspots; the sample period is a tunable parameter.

TABLE VII
MAHONY FILTER ON THREE CORES, SAME PIPELINE (MEASURED), WITH
THE ESKF SCALING ROW.

	text [B]	insns/update	CPU @ 20 MHz, 200 Hz
M7 fpv5	1780	354 / 390	(600 MHz target)
M4F fpv4-sp	1804	360 / 396	0.69% / 0.76%
M3 softfloat	2588	6187 / 8285	7.7% / 10.2%
ESKF (same pipeline)	63,310	~2.69M worst step (comparability era)	

Remaining hotspots ranked by measured instructions: the Simpson noise integral (162k, indexing overhead), the correction specialization (135k), the transition polynomial (130k, load-bound), the strided kernel (126k), copy read-backs (46k), guard compares (24k). Toolchain context: on the PX4 EKF2 covariance-prediction kernel, generated C from a mechanical Modelica port sits within 6% of upstream SymForce C++ on flash, instructions, and FP per call, on the same target and flags; the recorded caveat carries: the port transcribes SymForce’s *output*, so the comparison measures backend parity, not modeling-language overhead. Measurement methodology warning that must survive any re-measurement: the two-pass contraction split moved MACs into a second translation unit; count both or the step is undercounted.

XVII. A SMALL-PROCESSOR BASELINE: THE MAHONY COMPLEMENTARY FILTER

The same Modelica-to-C pipeline that produces the ESKF scales down to the smallest deployable attitude filter, and the pair brackets the toolchain’s reach. A clean-room implementation of the explicit complementary filter with gyro-bias feedback of Mahony, Hamel, and Pflimlin [13] was ported to the same library discipline (`modelica_models` branch `mahony-filter`): accelerometer correction gated on a 0.6 to 1.4 g plausibility band, parameter-gated magnetometer heading correction with a unit-aware epsilon, reciprocal-multiply quaternion renormalization, all scalar, in one equation-form when `sample` clause.

Validation (demonstrated: Tests. MahonyFilterStudy, 200 Hz, 30 s; OpenModelica 1.27.0 matches the rumoca compilation on all 12 reported error channels to 6 significant digits): a 170° initial attitude error with magnetometer aiding recovers to 3.67×10^{-3} rad; a 120° accelerometer-only tilt error to 1.4×10^{-7} ; under 25 Hz coning with 2 m/s² phase-locked vibration the steady tilt error is ~0.96 mrad with a documented rectified yaw drift of ~2.8 mrad/s absent heading aiding (the coning rectification a complementary filter cannot observe); a gyro bias of {0.01, -0.02, 0.015} rad/s is recovered to 3.9×10^{-4} rad/s.

Cost (measured: arm-gcc 15.2.1 -Os at the dossier flags, exact QEMU translation-block instruction counts over 500-step deltas including ~27 instructions of driver overhead; CPU percentages from a stated in-order cycle model at 20 MHz):

Table VII reports per-core figures (accel-only / with magnetometer); at 1 kHz the M4F carries the filter at 3.46% (3.80% with magnetometer) and even the softfloat M3 at

38.5% (51.0%). The whole linked demo including newlib `sqrtf` is 2.4 KB on the FPU cores and 4.9 KB on the M3; RAM (state 280 B, stack 96 B) is under 0.13% of a 320 KB part. The ESKF row uses the pre-optimization comparability numbers its package annotation carries (63,310 B text, ~2.69M instructions worst step); the current optimized step is the separately measured 954,168-instruction figure of Section XVI: the two eras are not mixed within a row.

[REVIEW] Three items the port surfaced. (a) A cross-tool when sample semantics trap: algorithm-form clocked blocks are split by OpenModelica into per-variable equations that read one-tick-stale values, while equation-form clauses agree across tools; the port uses equation form, but `Estimation/ComplementaryAttitude.mo` carries the latent hazard and is flagged in the campaign handoff. (b) The validation study is deterministic by repo pattern: no stochastic-noise case exists for this filter. (c) About 25 to 30 percent of the FPU update is wrapper and `sqrtf-chain` glue, being addressed by codegen work; the measured numbers include it honestly.

XVIII. RELATION TO THE CONING AND SCULLING ALGORITHMS THAT FLY

Production autopilots already carry a two-sample coning correction. This section shows that the derived bracket *is* that correction, that the algebra additionally supplies the sculling and scrolling terms and the next order without a new derivation per quantity, and, using the proved T^5 residual, that the shipped truncation order is provably sufficient at flight rates. Sources are cited by file and line as vendored today.

A. The shipped algorithms

PX4 accumulates every gyro sample trapezoidally, $\Delta\alpha = (\omega_k + \omega_{k+1})h/2$ (`src/modules/sensors/Integrator.hpp`, line 135), which is exactly the first-order-hold increment: the trapezoid mean of (16). Its coning accumulator (line 176) is

$$\beta += \frac{1}{2} \left(\alpha + \frac{1}{6} \Delta\alpha_{\text{prev}} \right) \times \Delta\alpha, \quad (48)$$

with α the angle accumulated so far. ArduPilot computes the same form (`libraries/AP_InertialSensor/AP_InertialSensor_Backend.cpp`, lines 340 to 348, citing Tian et al. [3]). Betaflight runs a complementary filter at PID rate and processes every sample, so it has no accumulation problem to solve.

B. Equivalence of the coning term

Proposition 9 (the shipped coning term is the bracket’s $\mathfrak{so}(3)$ component). *Split (48) into its two parts. The $\frac{1}{2}\alpha \times \Delta\alpha$ part is the composition (Baker-Campbell-Hausdorff) term that arises because those integrators accumulate a single rotation vector; the framework here composes group elements instead, so that part is performed exactly, to all orders, by the group product and is not an algorithm step (Proposition 1). The remaining part, $\frac{1}{12}\Delta\alpha_{\text{prev}} \times \Delta\alpha$, equals the $\mathfrak{so}(3)$ component $\frac{T^2}{12} \omega_k \times \Delta\omega$ of (15) to first order in the sampling parameter.*

Proof. Strategy: expand both stencils about the interval. Write $\omega_k = \omega$, $\omega_{k+1} = \omega + d$, $\omega_{k-1} = \omega - d$ (uniform slope). Then $\Delta\alpha_{\text{prev}} = (2\omega - d)h/2$, $\Delta\alpha = (2\omega + d)h/2$, and

$$\frac{1}{12}\Delta\alpha_{\text{prev}} \times \Delta\alpha = \frac{h^2}{48}(2\omega - d) \times (2\omega + d) = \frac{h^2}{12}\omega \times d,$$

which is exactly (15)'s $\mathfrak{so}(3)$ component with $\Delta\omega = d$. The coefficient $1/12$ is identical, not merely proportional. The two differ only in which finite difference estimates d : the shipped form uses a backward difference across the interval boundary, (15) a forward difference inside the interval, so the leading discrepancy is the local curvature term, first order in $2\pi fh$. ■

Verified numerically (Experiment 13): the two stencils differ by 0.30% of the exact per-interval correction at the deployed point, halving with each halving of h (first order, as proved), and by 0.008% at the 5.76 kHz reference. Two independent derivations, one from a Magnus expansion on the mixed-invariant group, the other from the Bortz-Savage-Tian strapdown route [7], [8], [3], converge on the same coefficient. That convergence is itself the strongest available check on both.

C. What the algebra adds: unification and no spectral assumption

In the strapdown lineage, coning and sculling are separately derived algorithms ([8] and [4] are literally separate papers, attitude and then velocity/position), each with its own coefficient set. Here one bracket produces all three at the same order: coning in $\mathfrak{so}(3)$, sculling in the velocity column, and scrolling through the time-block corner (Proposition 2, (15)), and the same machinery generates the next order mechanically (Lemma 3) rather than requiring a fresh derivation per kinematic quantity. Second, the coefficients come from the free Lie algebra with *no* assumption about the input spectrum, whereas Savage-class multi-sample coefficient sets are commonly obtained by fitting to an assumed sinusoidal coning input; the group-derived coefficients are the polynomial-fit class exactly and by construction (Section VIII). That is the prescriptive advantage: the correction is derived, not tuned to an assumed vibration model.

D. The higher-order terms as a sufficiency proof

The proved T^5 residual of Lemma 3 turns into a statement the classical route cannot make, because it lacks the exact residual: it *bounds* the shipped algorithm rather than replacing it.

Corollary 3 (sufficiency of the two-sample form at flight rates). *At the deployed configuration ($h = 1/1600$ s, $\|\omega\|$ up to 10 rad/s, vibration at the 200 Hz filter corner), the neglected T^5 terms are*

$$\frac{\|\Xi_5\| T^5}{\left\|\frac{T^3}{12}[N_0, N_1]\right\|} = \begin{cases} 1.77 \times 10^{-5} & \phi = 0.5 \text{ mrad}, \\ 6.60 \times 10^{-5} & \phi = 2 \text{ mrad}, \end{cases}$$

with Ξ_5 the proved grade-5 exponent of Lemma 3; the measured truncation error ratio agrees to three digits ($1.773 \times$

10^{-5} , 6.606×10^{-5} ; Experiment 12). Carrying the T^5 terms would first repay a 1% correction only below $h = 14.9$ ms ($f_s = 67$ Hz) at 0.5 mrad or $h = 7.7$ ms (130 Hz) at 2 mrad, or at body rates above 1252 rad/s (71,757°/s) at the deployed h .

Neither crossover is reachable by an airframe. The conclusion, stated plainly: *the two-sample coning form that PX4 and ArduPilot ship is truncated at the right order, with a computable margin of four to five orders*, and the higher-order terms of this paper are what bound that claim rather than what anyone should implement. Verified numerically; the bracket coefficients and the residual form it rests on are proved.

E. The honest qualification: which error actually binds

Corollary 3 bounds the *Magnus truncation* given the hold model. It does not say the hold model is adequate, and at the deployed configuration it is not the binding term. Experiment 13 measures, per interval at $2\pi fh = 0.785$, an exact correction to the trapezoid increment of 8.0×10^{-5} rad, of which the coning term (either stencil) accounts for only 0.8%: the remaining 99% is first-order-hold *interpolation* error against the true 200 Hz waveform, exactly the leg Theorem 2 bounds and the crossover of Fig. 5 exposes. The ratio is scale-invariant in h for a fixed vibration frequency (both legs are $O(h^3)$ per interval), so sampling faster does not change which term dominates; only a higher-order hold does. This is the same conclusion the crossover analysis reached from the other direction, and it is why the recommendation at 1600 Hz is the centered quadratic hold rather than more bracket terms. Measured.

F. Exact propagation versus bolt-on correction, and its cost

The equivalence above concerns the *correction*. The propagation it sits in front of is where the two architectures part. Both shipped stacks apply the corrected delta angle through a quaternion exponential, rotate the corrected delta velocity into the world frame, add gravity times the interval separately, and advance position by a first-order update; the velocity-to-position coupling and the gravity contribution are thus approximated per interval. The closed form obtains both exactly (Proposition 1): gravity is a left factor, the body input a right factor, and the coupling comes out of the nilpotent time block. The corrections of this paper are added to the *exponent* of that closed form ((16)), so a corrected step is still one exact flow evaluation, where a numerical integrator reaching the same order needs several evaluations per step.

Table VIII measures the three architectures propagating the same interval with an exact operation counter (Experiment 14: every arithmetic operation and transcendental call is counted by an instrumented scalar type, so the counts are measured, not estimated; accuracy is against a fine kinematic RK4 truth).

Three readings, all measured. (i) *Against a numerical integrator at matched accuracy the closed form wins, and the published figure is reproduced once the formulation is pinned.* On smooth motion (b) and (c) agree to four digits at every interval length tested, and (c) costs 284 flops

TABLE VIII
COST AT MATCHED ACCURACY, PER INTERVAL, MEASURED BY
OPERATION COUNTER, WITH END-OF-RUN ERROR OVER 1 S. SMOOTH
MOTION ISOLATES THE COUPLING; THE VIBRATION CASE IS
INTERPOLATION-DOMINATED.

architecture	flops	transc.	vel err	pos err
<i>smooth motion, $h = 10$ ms</i>				
(a) deployed bolt-on pattern	143	3	9.96e-3	7.10e-3
(b) RK4, four evaluations	396	1	9.72e-5	3.34e-5
(c) closed form + bracket	284	4	9.72e-5	3.34e-5
<i>smooth motion, $h = 50$ ms</i>				
(a) deployed bolt-on pattern	143	3	4.96e-2	3.64e-2
(b) RK4, four evaluations	396	1	2.44e-3	8.36e-4
(c) closed form + bracket	284	4	2.43e-3	8.33e-4
<i>200 Hz vibration, $h = 625$ μs (deployed)</i>				
(a) deployed bolt-on pattern	143	3	2.11e-3	1.03e-3
(b) RK4, four evaluations	396	1	2.14e-3	1.05e-3
(c) closed form + bracket	284	4	2.14e-3	1.05e-3

TABLE IX
EQUATION-GRAPH OPERATION COUNTS FOR THE SAME MIXED VECTOR
FIELD UNDER THREE FORMULATIONS, AGAINST THE PUBLISHED FIGURE
AND THIS PAPER’S EXECUTED-FLOP MEASUREMENT.

formulation	closed form	RK-4	ratio
general (ω_r , B symbolic)	552	784	1.42
$\omega_r = 0$ (standard INS)	322	568	1.76
$\omega_r = 0$, B literal	233	469	2.01
[1] Table I	360	677	1.88
this paper, executed scalar flops	284	396	1.39

against 396, a factor 1.39. The published closed-form paper [1] reports a 45% reduction, 677 against 360 operations per integration step (its Table I), a factor 1.88. A companion study settles the relation between the two by rebuilding an RK-4 on the *same* mixed vector field in CasADi and counting equation-graph operations, the instrument class of that Table I (Table IX, all rows measured). The study, its CasADi sources, and its rig are recorded alongside this paper (dev/2026-08-25-expmixed-codegen-study.md).

Two conclusions follow, and they replace what would otherwise be a hand wave. First, *the published 1.88 is bracketed by the $\omega_r = 0$ rows, 1.76 and 2.01, and its absolute counts 360 and 677 sit between those two specialized columns: Table I is consistent with the standard-INS specialization, in which the right algebra element carries no rotation and the entire right-hand rotation machinery vanishes structurally. That identifies which formulation the published figure corresponds to. Second, the two instruments do not disagree at all: the general-formulation graph ratio is 1.42 and this paper’s executed-flop ratio is 1.39, agreeing to within 2%. The whole apparent discrepancy was formulation scope, not counting method. For completeness, a matched-shape run of this paper’s own instrument, zero-order hold on both sides against the same kinematic RK-4, gives 221 against 381 flops, a factor 1.72 (Experiment 15).*

(i.b) On the flight target, and what the instruction counts do and do not show. The same study reports executed Cortex-M7

instructions for one step of each architecture at the same input point (arm-none-eabi -Os, cortex-m7 fpv5-d16, -fno-math-errno): 880 for the closed form, plus roughly 180 estimated instructions inside libm, against 1,132 for RK-4 on the same field. These two rows are not at matched accuracy: one RK-4 step over a 0.6 rad rotation is wrong in the third significant figure, so holding a tolerance would mean subdividing at a cost linear in the substeps while the closed form is step-size independent. The residual instruction advantage at one step each is therefore a floor on the closed form’s advantage, not its value. What the measurement does retire is this paper’s own earlier worry that transcendental calls might reverse the sign: they cannot, because newlib’s `sinf` and `cosf` fast path costs 22 to 23 instructions on this argument range, not the hundreds a generic argument reduction would. (i.c) The advantage grows with the interval. [1] notes that an RK-4 step must be subdivided to hold a tolerance as the step grows, at a cost linear in the added substeps, while the closed form is step-size independent. That is the same scoping this paper reaches from the accuracy side in reading (iii): the exactness argument lives at packet-rate propagation and coarser, which is precisely where a preintegrating estimator operates. (ii) The exactness of the coupling is worth two orders where the interval is long enough to expose it. At a 10 ms packet interval on smooth motion the deployed pattern is $103\times$ worse in velocity and $212\times$ worse in position than either exact-propagation variant; at 50 ms, $20\times$ and $44\times$. This is precisely the coupling that a first-order velocity and position update approximates and the time block obtains exactly. (iii) At the deployed per-sample rate all three land in the same accuracy class. With 200 Hz vibration at $h = 625$ μ s the three agree within 2% of each other, because the error there is interpolation-dominated (Section XVIII), not coupling-dominated. The exactness argument is therefore an argument about packet-rate and coarser propagation, which is exactly where a preintegrating estimator operates, and not a claim that the deployed per-sample pattern is inaccurate at its own rate.

G. What remains genuinely new for implementation

Two items, neither of which is a coning coefficient. (a) *Bias Jacobians of the correction terms to matching order* (Proposition 4): production preintegration carries the coning and sculling corrections but not their sensitivity to the bias estimate at the same order, so a bias update moves the corrected increments inconsistently with the correction that produced them; the $\frac{T^2}{12}$ Jacobian terms, including the velocity-to-gyro-bias cross term with no ZOH counterpart, close that. (b) *The centered quadratic hold* (Proposition 6): a sensor-backend accumulator is necessarily causal, so no production integrator can use a stencil that reaches past the interval; the delayed fusion horizon (Remark 6) licenses exactly that for the fused increments at zero added lag. This is the one place where the *architecture*, not the algebra, is the differentiator, and by Section VII it is worth $5\times$ to $6\times$ at the deployed rate, against a coning refinement worth nothing there.

TABLE X
FAIRNESS-AUDITED STATIONARY-PAD REPLAY, 3D POSITION RMSE
(IDENTICAL SENSOR STREAMS; MEASURED).

window	generated	PX4 EKF2	AP EKF3
pad, vendor defaults	0.076 m	0.285 m	0.087 m
30 s [13, 43] s	0.083 m	0.234 m	0.181 m

XIX. ACCURACY VALIDATION: SIMULATION AND REPLAY

[REVIEW] Scoping, honored in the title and abstract: the sources contain *no flight-hardware logs*; every accuracy number in this paper is closed-loop simulation or standalone replay of simulator-generated streams on the compiled artifact. The hardware campaign that would support a flight-validation claim is an open item (Section XX).

A. Mission-level closed loop

45 s box missions (rumoca sim, 5 ms estimator tick; trajectories preserved at a 0.5 s grid): GPS-navigation mission nav-vs-truth 3D RMSE 0.149 m (max 0.59 m); local mode 0.137 m (max 0.52 m); flown-trajectory match between the modalities 0.195 m RMSE, within the documented navigation-modality tolerance. The ZOH-vs-FOH mission row (0.148753 vs 0.148903 m, no measurable change on this benign aided mission) already appears in the Part I flight-estimator study (Section VII) and is not restated as a separate result. All measured.

B. The EKF2/EKF3 comparison and the fairness audit

Round 2 (three filters replayed on one shared flattened sensor stream from a 5 s GPS-aided flight): the compiled NavigationEstimator replayed *standalone* matches the in-sim ESKF to 4% (0.150 vs 0.143 m), closing the in-sim-vs-replay asymmetry; vendor-default 3D position RMSE 0.150 m (generated), 0.255 m (PX4 EKF2), 0.364 m (ArduPilot EKF3). Round 2.5 audited the reviewer hypothesis that the EKF3 ranking was a methodology artifact, and it was: a 12 s synthetic noise-free warm-up froze a ~ 0.34 m horizontal datum offset and mis-seeded EKF3’s baro bias. On a genuine simulator-generated stationary pad with each stack’s natural alignment (flight window [13, 18] s):

The EKF2-vs-EKF3 ranking *reverses* on the fair pad (Table X). Findings that survive the audit: EKF3’s declination yaw bias (fixable) and its velocity tracking on the aggressive box run; EKF2’s vertical carries the 0.4 m baro bias nearly undiminished. The generated estimator is unaffected by all four audit fixes and is the top filter on the fair pad; the carried caveat: it is favorably matched to the simulator tuning (it is the simulator’s own filter configuration). The audit narrative is itself a contribution: it documents, with commands, how replay-benchmark rankings are manufactured by warm-up protocol.

C. Consistency: NEES and NIS

Closing the ledger’s largest validation gap for the simulation domain: filter consistency, measured on both qualification missions. Method: the estimator was instrumented with

add-only outputs mirroring the full 15-state error covariance and the bias estimates per tick (modelica_models branch nis-nees-instrumentation); the per-tick NIS, correction source, and per-source acceptance flags were already published on the status record. The error is formed in the convention of Lemma 6: $\varepsilon_p = \hat{R}^\top(p_{\text{true}} - \hat{p})$, likewise velocity, $\varepsilon_\theta = \text{Log}(\hat{R}^\top R_{\text{true}})$, and bias errors against the plant’s true random-walked biases (the simulation owns them); $\text{NEES} = \varepsilon^\top P^{-1} \varepsilon$. Both missions parameterize their sensor-noise seed, so a five-seed ensemble per mission was run (45 s each, the qualification settings) and the average NEES (ANEES) is plotted against the 95% ensemble chi-square bounds ($N = 5$: [1.25, 5.50] for 3-dof blocks, [10.59, 20.17] for the 15-state total); NIS is shown for the representative first seed with single-run bounds per source dof, the GPS dof inferred per tick from the joint acceptance flags. Data: data/nees_{gps,flow}.dat, data/nis_{gps,flow}*.dat; regeneration: Vehicles/Rdd2/Test/ConsistencyEnsemble.mo wrappers through rumoca sim, post-processed by the harness consistency binary (quote-aware extraction; bulk CSVs deleted after extraction). All measured, simulation.

Verdict, carried to the ledger: the covariance pipeline is calibrated on the translational channels and on every NIS channel except a mildly optimistic magnetometer; attitude covariance is optimistic by about $2\times$, concentrated in the aggressive descent-and-landing phase. This is a tuning-level finding (attitude and heading noise parameters), not a structure-level one: the transitions, resets, and gates behave as designed throughout, including the gated rejection of the post-outage GPS re-acquisition sample. All measured, simulation.

[REVIEW] The attitude-covariance optimism ($2.2\times$ ANEES, landing-phase concentrated) and the mildly optimistic magnetometer NIS are consistency findings this paper reports but does not fix: the retuning decision (gyro or heading process/measurement noise, or a landing-phase-aware inflation) is open, and the tangent-group gate experiment of Section IX-B should reuse this harness.

D. What is not validated

Claims explicitly *not* carried by this paper: no flight-hardware data exists (simulation consistency is now measured, Section XIX-C, with its attitude finding recorded rather than fixed). The hardware campaign goes to the open items with effort estimates. The optical-flow-only mission and swappable-interface qualification live on an unmerged branch; merge status must be re-checked at publication time.

XX. KNOWN LIMITATIONS AND OPEN ITEMS

The consolidated ledger, grouped, each item with status and effort; this section absorbs the Part II open items and joins the Part I review checklist as the paper’s honesty spine.

Intrinsic (provable, not fixable by architecture). (1) Bias-transport impossibility: no group action that forgets the input path can transport bias corrections exactly (Section VI-C); the $O(\|db\|^2)$ remainder is a property of the problem, evaluating

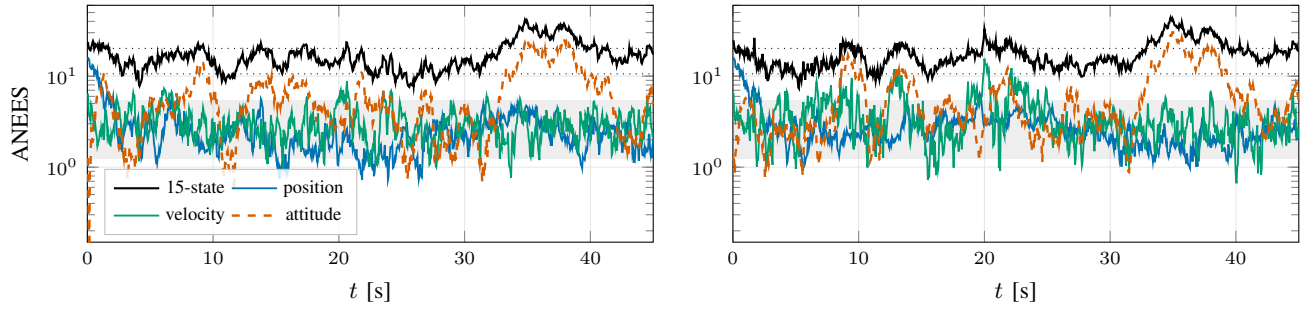


Fig. 6. ANEES, five-seed ensembles. Left: GPS mission; right: GPS-denied optical-flow mission (outage 10 to 20 s). Shaded band: 95% ensemble bounds for the 3-dof blocks; dotted lines: bounds for the 15-state total. Verdict, plainly: position (mean 2.64/2.85 against 3) and velocity (3.07/3.63) are *calibrated*, and position stays calibrated through the GPS-denied window (2.94) carried by flow aiding; attitude is *optimistic* on both missions (mean 6.47/6.40 against 3, about $2.2\times$), concentrated in the 35 to 40 s landing window (peaks 17.2/16.6) with 2.7 to 7.3 through cruise, and the 15-state total (17.3/17.8 against 15) inherits it. The initialization transient reflects the deliberate initial-position offset, a qualification witness, absorbed within the first second.

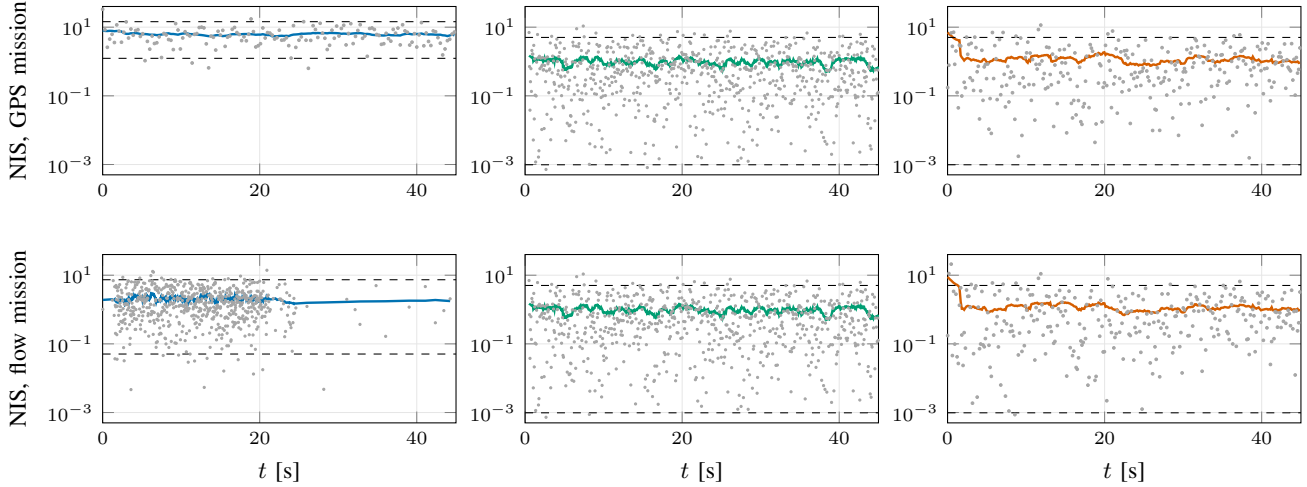


Fig. 7. Per-source NIS, representative seed, with single-run 95% chi-square bounds (dashed) and a 25-event running mean (colored). Top, GPS mission: GPS joint 6-dof (mean 6.27 against 6, near-calibrated), barometer 1-dof (0.98, calibrated), magnetometer 1-dof (1.32, mildly optimistic, consistent with the attitude finding of Fig. 6). Bottom, flow mission: optical flow 2-dof (2.05 against 2, calibrated), barometer (0.98), magnetometer (1.37). The flow-mission GPS channel (not shown) means 6.36 over its 35 aided seconds; at $t = 24.9$ s a single position-only re-acquisition attempt after the outage returned NIS 89.6 and was *rejected* by the gate: the supervision layer behaving as designed.

to $\sim 4 \times 10^{-11}$ relative at typical corrections: negligible, permanent, no action.

Approximation-level (bounded, upgradeable). (2) Retrodict single-sample hold over ages to 0.25 s: upgrade by buffering per-packet deltas and reapplying across the age window (exact by Theorem 6) or by FOH on the retrodiction pair; about a day plus a mission A/B, with an hours-level isolation measurement (zero-latency wiring diff) first. (3) Delay transport not mechanically proven: H_d is proven to 1.33×10^{-15} , the $\Phi(-\text{age})$ factor is block-derived only; one to two days to extend the proof harness. (4) Shared-endpoint noise correlation not modeled in Q_d (Section VI-D(ii) gives the bound); one to two days, relevant once FOH defaults on. (5) The $\varepsilon_\theta \times r_0$ second-order innovation term: standard neglect; an hours-level paragraph ties it under the gate.

Verification debt. (6) Ad-equivalence of the packet transition blocks (47) by block inspection, not mechanized; same for the UKF paths; about a day in the jacobian-proof rig. (7)

NEES/consistency validation: *resolved for simulation* (Section XIX-C: five-seed ANEES on both missions, translational channels and NIS calibrated, attitude optimistic by about $2\times$ in the landing phase, magnetometer NIS mildly optimistic). Remaining: the attitude/heading retuning decision the finding opens, hardware NEES, and the scripted outage-plus-misalignment gate variant that decides the tangent-group question (Section IX-B). (8) The Part I review-checklist items Part II inherits: per-fusion correction magnitudes from logs (hours); coning budget regeneration (hours); Savage comparison (a day); harness publication (decision). (9) Flight-hardware validation: none; weeks; gates any flight-validation claim (Section XIX). (10) An ESKF-vs-UKF head-to-head row (Section XV): half a day plus one benchmark run. (11) A first-order equivalence lemma between the transported-Jacobian and buffered-window fusion forms (Section XI): about a day.

Architectural precondition (new). (12) No sub-sample accumulation is implemented: the flight firmware consumes one

instantaneous IMU sample per estimator release and discards the rest, and the Modelica model holds a single sample per interval, so the composed sub-sample architecture every hold result here presupposes is intended (recorded by the preintegration-period parameter and the coning-corrected delta fields of the sample record) but not built (Section VIII, p. 16). Closing it, by accumulating each sample into a delta packet, is the precondition for any hold-order result to apply on hardware; until then the deployed integrator is one-sample-per-release ZOH. Effort: driver plus model change with an A/B against the existing missions.

(13) Executed-instruction counts on the flight target: *largely retired*. A companion codegen study measured one step of each architecture on Cortex-M7 (880 instructions for the closed form against 1,132 for RK-4 on the same field) and established that transcendental cost cannot reverse the sign, since the $\sin/\cos$ fast path costs 22 to 23 instructions on this argument range (Section XVIII-F). What remains is a *matched-accuracy* instruction comparison: the measured rows put RK-4 at one step per interval, where it is not converged, so the observed advantage is a floor. Effort: subdivide RK-4 to a fixed tolerance and re-measure, plus the deployed bolt-on pattern as a third row.

Tooling honesty. The checked-in scenario configs used a file form that failed under a since-fixed compiler error; re-check at publication time; the flat-model emitter dies on the mission model; neither affects the measured artifacts.

Deliberate scope decisions (documented, not defects). No tangent-group error state (Section IX-B, with its re-entry gate); terrain and barometer datum as auxiliary scalar filters rather than augmented states, and the range row unwired (Section XII); mocap without a latency model; 1 kHz as an ambition, not a spec, with the sample period a parameter and the measured margin statement of Section XVI.

XXI. CONCLUSION

This paper contributes an exact error theory, not a new integrator. The increment it analyzes is already deployed: PX4 and ArduPilot accumulate trapezoidally and apply a two-sample coning correction that Section XVIII identifies, coefficient for coefficient, with the $\mathfrak{so}(3)$ component of the derived bracket. What the classical route cannot supply, and what follows here from the free Lie algebra with no assumption on the input spectrum, is the exact residual: the T^4 grade vanishes identically and the T^5 term carries the proved coefficients $-1/240$ and $-1/720$. The headline consequence is a sufficiency statement about deployed practice rather than a proposal to change it: at flight rates the shipped two-sample form is provably adequate with a margin of four to five orders, and the crossovers where that ceases to hold (67 to 130 Hz sampling, or 1252 rad/s body rates) lie far outside any airframe. The same bound predicts the measured marginality of first-order hold at the deployed 1600 Hz rate instead of merely recording it.

The layering is worth stating once more, because it is easy to conflate. The exact mixed-invariant propagation, and the result

that it costs less than a Runge-Kutta step of the same problem, are the prior papers' [1], [2]. What this paper adds on that foundation is that the hold corrections enter the closed form as terms of its exponent, with proved coefficients and an exact residual, so a corrected step remains one exact flow evaluation: measured at matched accuracy, 284 flops per interval against 396 for a four-evaluation RK4, while the deployed bolt-on pattern, cheaper per interval, is two orders less accurate in velocity and position once the interval is long enough for the coupling to matter. The deployed correction is provably adequate, the propagation here is exact where the deployed one approximates, and at matched accuracy it is the cheaper of the two exact routes.

On that foundation the paper composes an end-to-end bound with explicit constants, in which bi-invariance and the 2-step nilpotent ideal remove the Gronwall factor that would otherwise make a per-second statement vacuous; an online certificate computed from interior-sample residuals; exact reapplication of buffered preintegrals through the state-independent factorization, with a second-order bias remainder that is intrinsic because exact bias transport is impossible in any framework; a centered quadratic hold whose novelty is architectural, since sensor-backend accumulation must be causal while delayed fusion licenses non-causal stencils, with its practical value open pending a flight-estimator A/B; and a unification in which coning, sculling, and scrolling are one bracket's three components, carried consistently into the bias Jacobians that production preintegration omits.

The rate decision the vehicle actually made sharpens what that buys. At the deployed 1600 Hz with motor vibration at the 200 Hz filter corner, $2\pi fh \approx 0.79$ sits inside the marginal band of this paper's own admissibility condition, so *plain first-order hold is not a clear win at the deployed configuration*: measured against a held-interval-start baseline it still gains $5\times$ to $7\times$, and measured against a held-interval-end baseline at the nearest study points it is at parity or slightly worse. The constructive answer is the centered quadratic hold, one order faster in h , measured a further $5\times$ to $6\times$ below FOH at the deployed point, and admissible at zero added lag only because fusion is delayed. That is the recommended configuration at 1600 Hz, and it is the clearest practical payoff of the delayed-fusion architecture here. Both statements are contingent on an architectural precondition this paper records rather than assumes: neither the flight firmware nor the model accumulates IMU samples between estimator releases today, so any hold-order result applies to the architecture the system is designed to become, not to the one running now. theory, the paper delivers an end-to-end guarantee: interpolation and flow-sensitivity bounds whose Gronwall factor the algebra eliminates compose into a provable per-second attitude bound with explicit constants, accompanied by an online a posteriori certificate that can ship inside the generated flight code. The delayed-fusion architecture is what makes the stronger stencils and the guarantee practical: centered, smoothing-grade holds are admissible at zero added lag for the fused state, and the mixed-invariant factorization turns horizon corrections into exact

reapplication of precomputed factors, with a proved second-order relinearization remainder that is negligible at realistic correction magnitudes. The bound now covers the full loop: interpolation plus truncation plus reapplication remainder. Numerical falsification against RK4 ground truth confirmed every derivation (orders 3.006, 5.008, and remainder slope 1.992 against theory 3, 5, 2), corrected one informal claim (the relative residual scales as $\frac{1}{10} \|\Delta\omega\| T + \frac{1}{15} (\|\omega\| T)^2$), and reported bound tightness honestly (phase-blind factors 389 to 4509 on worst-case coning signals). Part II closes the loop from theorem to artifact: the same closed forms drive a 15-state delayed-fusion ESKF whose measurement Jacobians are mechanically verified at the capture-time state, whose covariance bookkeeping is exact-sparse and PSD-preserving with measured counterexamples behind every supervision decision, and whose compiled single-precision artifact replays standalone to within 4 percent of its own simulation and tops a fairness-audited replay benchmark against PX4 EKF2 and ArduPilot EKF3, at 0.076 m pad RMSE and 0.149 m mission RMSE, all in simulation and replay, never yet in flight.

The relation to what flies is now explicit: the two-sample coning term that PX4 and ArduPilot ship is the bracket's $\mathfrak{so}(3)$ component with the identical $1/12$ coefficient, reached independently from the Bortz-Savage-Tian route, and the proved T^5 residual shows that truncation to be right by four to five orders at flight rates, with the crossovers where it would stop being right (67 to 130 Hz sampling, or 1252 rad/s body rates) far outside any airframe. The honest qualification travels with it: at the deployed configuration the binding per-interval error is not coning but hold interpolation, which outweighs the coning term by two orders and which only a higher-order hold removes.

The consolidated open-items ledger (Section XX) carries the honesty spine: the NEES calibration gap, the retrodiction hold upgrade, the delay-transport mechanization, and the hardware campaign that would earn a flight-validation claim. None of the open items affects the shipped correction at the stated rates.

APPENDIX

A. The extended-algebra commutator

For $\xi_i = \begin{bmatrix} \Omega_i & U_i \\ 0 & \beta_i B \end{bmatrix}$,

$$\xi_1 \xi_2 = \begin{bmatrix} \Omega_1 \Omega_2 & \Omega_1 U_2 + \beta_2 U_1 B \\ 0 & \beta_1 \beta_2 B^2 \end{bmatrix}, \quad B^2 = 0,$$

so $[\xi_1, \xi_2]$ is (14), closed in $\hat{\mathfrak{g}}$ with zero B component. With $U = [u_1 \ u_2]$, $UB = [0 \ u_1]$: the corner terms move first columns into second (position) columns, and a second application of B annihilates them, which is the mechanism behind both Proposition 2 and the single-corner claim.

B. Simplex moments

Over the ordered simplex $0 < t_3 < t_2 < t_1 < T$:

$$\int t_1 = \frac{T^4}{8}, \quad \int t_2 = \frac{T^4}{12}, \quad \int t_3 = \frac{T^4}{24}, \\ \int t_1 t_2 = \frac{T^5}{15}, \quad \int t_1 t_3 = \frac{T^5}{30}, \quad \int t_2 t_3 = \frac{T^5}{40}.$$

Hence $\int(t_3 - t_2) = -\frac{T^4}{24}$, $\int(t_1 - t_2) = +\frac{T^4}{24}$ (Lemma 2), and $\int t_1(t_3 - t_2) = -\frac{T^5}{30}$, $\int t_3(t_1 - t_2) = +\frac{T^5}{120}$, whose weighted sum $\frac{1}{6}(-\frac{T^5}{30} + \frac{T^5}{120}) = -\frac{T^5}{240}$ independently confirms the $-\frac{1}{240}$ coefficient of Lemma 3 by direct integral evaluation (Remark 2); the same moments with t_i^2 give the $\frac{1}{360}$ coefficient of Proposition 6.

C. The full mixed exponential

Listing 3 reproduces `exp_mixed` in full, unabridged and unchanged on the `foh-preintegration` branch; Listing 1 in the main text annotates its load-bearing lines. Experiment 2 of Section VII verifies this exact construction against the generic matrix exponential (8.7×10^{-19}) and against RK4 of the mixed flow (2.4×10^{-14}).

```

within LieGroups.SE23.Quat;
function exp_mixed "Mixed exponential for SE_2(3) INS
propagation"
  input Real X0[10] "Initial state {p0, v0, q0}";
  input Real l[9] "Left (body-frame) algebra element {vb, ab
, omega_l}";
  input Real r[9] "Right (world-frame) algebra element {vb_r
, ab_r, omega_r}";
  input Real B[2,2] "Coupling matrix (typically
{{0,1},{0,0}})";
  output Real X1[10] "Updated state {p1, v1, q1}";
protected
  Real omega_l[3], omega_r[3];
  Real theta_sq;
  Real C1, C2, C3;
  Real theta;
  // The closed coefficients are cancellation-prone below
  0.1 rad in
  // single-precision generated code; the retained series is
  accurate there.
  constant Real eps = 1e-2;

  // Omega matrix and its square
  Real Om[3,3], Om2[3,3];

  // A matrices (3x2): [a_b | v_b] for left and right
  Real Al[3,2], Ar[3,2];

  // N matrices (3x2): calculated for left and right
  Real Nl[3,2], Nr[3,2];

  Real I2[2,2];
  Real IpB[2,2];

  // Rotation quaternions
  Real q_l[4], q_r[4], q_r0[4], q1[4];
  Real R_r0[3,3], R_r[3,3];

  // P matrices (3x2): [v | p]
  Real P0[3,2], P1[3,2];
  Real term1[3,2], term2[3,2], term3[3,2];
algorithm
  omega_l := l[7:9];
  omega_r := r[7:9];

  theta_sq := omega_l[1]^2 + omega_l[2]^2 + omega_l[3]^2;

  if theta_sq < eps then
    C1 := 0.5 - theta_sq / 24.0;
    C2 := 1.0/6.0 - theta_sq / 120.0;
    C3 := 1.0/24.0 - theta_sq / 720.0;
  else
    theta := sqrt(theta_sq);
    C1 := (1.0 - cos(theta)) / theta_sq;
    C2 := (theta - sin(theta)) / (theta_sq * theta);
    // C3 = (theta^2/2 + cos(theta) - 1) / theta^4
    C3 := (theta_sq/2.0 + cos(theta) - 1.0) / (theta_sq *
theta_sq);
  end if;

  // Build Omega (skew of omega_l) and Omega^2.
  Om := LieGroups.SO3.Quat.wedge(omega_l);

```

```

Om2 := Om * Om;

// A_l = [a_b | v_b] (columns: acceleration, velocity from
// left algebra)
A1 := {{l[4], l[1]}, {l[5], l[2]}, {l[6], l[3]}};
Ar := {{r[4], r[1]}, {r[5], r[2]}, {r[6], r[3]}};

I2 := identity(2);

// N_l = A + A*B/2 + Om*A*(C1*I + C2*B) + Om^2*A*(C2*I +
// C3*B)
Nl := A1 + 0.5 * A1 * B
+ Om * A1 * (C1 * I2 + C2 * B)
+ Om2 * A1 * (C2 * I2 + C3 * B);

// N_r: same closed-form construction with right algebra
// and -B.
// The Omega in Nr uses omega_r.
// For the standard INS case, r has omega_r = 0, so Om_r =
// 0 and Nr = Ar + Ar*(-B)/2

// For Nr, we need omega_r's series coefficients
theta_sq := omega_r[1]^2 + omega_r[2]^2 + omega_r[3]^2;
if theta_sq < eps then
  C1 := 0.5 - theta_sq / 24.0;
  C2 := 1.0/6.0 - theta_sq / 120.0;
  C3 := 1.0/24.0 - theta_sq / 720.0;
else
  theta := sqrt(theta_sq);
  C1 := (1.0 - cos(theta)) / theta_sq;
  C2 := (theta - sin(theta)) / (theta_sq * theta);
  C3 := (theta_sq/2.0 + cos(theta) - 1.0) / (theta_sq *
  theta_sq);
end if;

Om := LieGroups.SO3.Quat.wedge(omega_r);
Om2 := Om * Om;
Nr := Ar - 0.5 * Ar * B
+ Om * Ar * (C1 * I2 - C2 * B)
+ Om2 * Ar * (C2 * I2 - C3 * B);

// Rotation updates
q_l := LieGroups.SO3.Quat.exp_map(omega_l);
q_r := LieGroups.SO3.Quat.exp_map(omega_r);
q_r0 := LieGroups.SO3.Quat.product(q_r, X0[7:10]);
q_l := LieGroups.SO3.Quat.product(q_r0, q_l);

R_r0 := LieGroups.SO3.Quat.to_DCM(q_r0);
R_r := LieGroups.SO3.Quat.to_DCM(q_r);

// P0 = [v0 | p0]
P0 := {{X0[4], X0[1]}, {X0[5], X0[2]}, {X0[6], X0[3]}};

// I + B
IpB := I2 + B;

// P1 = R_r0 * Nl + (R_r * P0 + Nr) * (I + B)
term1 := R_r0 * Nl;
term2 := R_r * P0 + Nr;
term3 := term2 * IpB;
P1 := term1 + term3;

// Extract p = P1 column 2 and v = P1 column 1 to match {p
// , v, q} ordering.
Xl[1] := P1[1,2]; Xl[2] := P1[2,2]; Xl[3] := P1[3,2];
Xl[4] := P1[1,1]; Xl[5] := P1[2,1]; Xl[6] := P1[3,1];
Xl[7] := q1[1]; Xl[8] := q1[2]; Xl[9] := q1[3]; Xl[10] :=
q1[4];
end exp_mixed;

```

Listing 3. LieGroups/SE23/Quat/exp_mixed.mo, complete (modelica_models branch foh-preintegration).

D. Numerical tables

Tables superseded in the main text by Figs. 1, 2, and 5; kept here so the numbers stay citable.

ACKNOWLEDGMENT

[FILL IN: Acknowledgment and funding text; the precursor paper acknowledged NXP and the CogniPilot

TABLE XI

FOH TRUNCATION ERROR AGAINST RK4 TRUTH (FROBENIUS). ERR(I): TRAPEZOID ONLY; ERR(II): WITH $+\frac{T^3}{12}[N_0, N_1]$; SIGN: WITH $-\frac{T^3}{12}$; PRED: $\|T^5\Xi_5\|$ FROM LEMMA 3.

case	T [s]	err(i)	err(ii)	sign	err(ii)/pred
F1 ($\Delta\omega=0.2, \Delta a=2$)	178e-6	2.51e-8	8.25e-14	5.03e-8	1.003
F2 ($\Delta\omega=0.5, \Delta a=5$)	1e-3	3.26e-6	1.07e-10	6.51e-6	1.000
S1 ($\Delta\omega=2, \Delta a=10$)	1e-3	7.60e-6	7.24e-10	1.52e-5	1.000
S2 ($\Delta\omega=2, \Delta a=10$)	1e-2	6.94e-4	1.05e-6	1.39e-3	0.998

TABLE XII

PER-SECOND ATTITUDE ERROR: A PRIORI BOUND VS MEASURED, PER-SAMPLE INTEGRATORS ON BAND-LIMITED CONING SIGNALS (RAD OVER 1 S; SEED 31415; PLOTTED PER-TICK SINGLE-LINE DATA IN FIG. 2).

f [Hz]	ϕ [mrad]	FOH			centered quadratic		
		bound	meas.	ratio	bound	meas.	ratio
100	0.5	4.7e-4	1.2e-6	389	2.6e-5	4.8e-7	54
200	0.5	3.7e-3	4.0e-6	924	4.2e-4	6.1e-7	684
400	0.5	3.0e-2	1.8e-5	1641	6.7e-3	1.5e-6	4509
100	2.0	1.9e-3	4.6e-6	408	1.0e-4	2.9e-7	362
200	2.0	1.5e-2	2.5e-5	593	1.7e-3	4.2e-7	3943
400	2.0	1.2e-1	1.7e-4	706	2.7e-2	8.4e-6	3179

Foundation; confirm for this paper.]

[REVIEW] Length: with the end-to-end bounds (Section IV), the delayed-fusion holds (Section V), and the reapplication section (Section VI), the paper exceeds a typical 6-page L-CSS or 8-page conference limit. The bulk numeric tables (Tables XI and XII) are already demoted to the appendix in favor of Figs. 1–5, and the full `exp_mixed` listing sits in Appendix C with only the abridged, annotated listings in the text (the readable-specification thesis earns them their place). Remaining candidates to move rather than cut: the coefficient-matching displays of Lemma 3(iv) and the proofs of Theorems 2 and 5. Thirty-one pages as of the cost-comparison cycle; the recorded future split lines (comment block at the top of the source, and the document-role statement) are the intended resolution rather than cuts.

REVIEW CHECKLIST

Every review box marked in the text, with its page; the consolidated open-items ledger with statuses and efforts is Section XX:

- 1) Author list, order, and affiliations verified against the published paper; e-mail addresses, corresponding author, and authorship of this extension remain (p. 1).
- 2) Published paper obtained and cited: title, authors, and date verified from the record; only volume, pages, and DOI remain to be filled from the IEEE listing (`refs.bib`).
- 3) Notation alignment with the published L-CSS version; the arXiv precursor's Theorem 1 typos noted (p. 3).
- 4) Bias-slope invariance rests on the sensor-filtering assumption; the specific part's filter chain is unaudited (p. 7).
- 5) Bias-Jacobian recursions: velocity channel FD-verified in the adversarial review, and all five channels, position included, FD-verified in the flight-estimator implementation; all channels now exercised (p. 7).

- 6) Publication venue and link for the falsification harness (p. 15).
- 7) Savage five/six-sample coefficient comparison: open (p. 16).
- 8) Error-budget numbers to be re-derived by the simulation harness (p. 17).
- 9) Uniform, jitter-free, synchronized sample timestamps assumed by the FOH slope, the centered stencils, and the a posteriori certificate; driver behavior unaudited (p. 9).
- 10) Delay-buffer alignment and the 200 ms fusion-horizon architecture: taken from the system design context, not read from a single implementation site (p. 9).
- 11) Rate-feedback path audited in the model and confirmed on the deployed stack: ICM-45686 at 1600 Hz ODR with 200 Hz silicon gyro LPF, data-ready-paced rate loop; silicon-LPF coning attenuation noted as an uncredited observation, and RPM-synced notching noted as future driver work (p. 9).
- 12) Typical per-fusion bias-correction magnitudes assumed, not grounded in the implementation's tuning (p. 11).
- 13) Error-state convention consistency: ESKF path read directly, UKF path audited by the implementation review with no disagreement found; remaining item is the block-inspection status of the adjoint-equivalence (p. 11).
- 14) Tangent-group numerics of the Galilean-group remark come from the companion analysis harness in a session scratchpad; fold into `foh-check` for reproducibility (p. 12).
- 15) Covariance-side omissions recorded (colored-noise recursion, full predict/correct treatment), deferred to a companion architecture document in preparation (p. 12).
- 16) Mahony port items: the cross-tool when-sample semantics trap (latent in `ComplementaryAttitude.mo`), no stochastic-noise study case, and the measured wrapper glue share of the FPU update (p. 23).
- 17) Consistency findings reported, not fixed: attitude ANEES about 2.2x with landing-phase concentration and mildly optimistic magnetometer NIS; retuning decision open (p. 26).
- 18) Title, abstract, and validation scoped to simulation and replay; flight-hardware campaign gates any flight-validation claim (p. 26).
- 19) UKF section is an explicitly marked outline; head-to-head benchmark row open (p. 22).
- 20) Pin the propagation-step citations for both stacks (PX4 EKF2 predict step; ArduPilot AP_NavEKF3 strapdown update) to a commit, alongside the accumulator line numbers already given (p. 23).
- 21) Recover the exact Tian et al. 2010 citation fields from the ArduPilot source comment; PX4 and ArduPilot line numbers are as vendored today and should be pinned to a commit (p. 23).
- 22) Deployed-rate scope: design point restated at 1600 Hz, the 5.76 kHz mode retained only as a measured reference; FOH marginal at the deployed point and the

centered hold recommended instead (p. 17).

- 23) Architectural gap: neither firmware nor model accumulates sub-samples today, so hold-order results describe the intended architecture (p. 16).
- 24) Venue page limit exceeded; appendix-move candidates listed (p. 30).

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